Electronic supplementary material for

The predictive value of myoglobin for COVID-19-related adverse outcomes: a systematic review and meta-analysis

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References

Abbreviations: Mb, myoglobin; cTnI, cardiac troponin I; ICU, intensive care unit**.**

Methods S1. Definition

1. Laboratory-confirmed case

Laboratory-confirmed case was defined as the presence of SARS-CoV-2 in respiratory specimens (including nasal and pharyngeal swabs) detected by the reverse-transcriptase polymerase chain reaction (RT-PCR), which was conducted in accordance with the protocol established by the World Health Organization.[1] Practice of the diagnostic criteria was based on the recommendation by the National Institute for Viral Disease Control and Prevention of CDC (Available at http://ivdc.chinacdc.cn/kyjz/202001/t20200121_211337.html).

2. Severe disease

Severe disease was defined as meeting one of the following criteria: 1) presence of shortness of breath with a respiratory rate ≥ 30 breaths/minute; 2) an oxygen saturation (SpO2) ≤ 93% in the resting state; 3) hypoxemia defined as an arterial

partial pressure of oxygen divided by the fraction of inspired oxygen (PaO2/FiO2 ratio) ≤ 300 mmHg; 4) presence of radiographic progression, defined as ≥ 50% increase of target lesion within 24-48 hours. The criterion was based on the Chinese COVID-19 prevention and control program (6th edition, http://www.nhc.gov.cn/yzygj/s7653p/202002/8334a8326dd94d329df351d7da8aefc2.shtml, accessed Feb 18, 2020) and American Thoracic Society guideline[2].

Methods S2. Search strategies

Search strategy for PubMed (September 8, 2021)

#	Searches	Results
#1	((covid-19[Title/Abstract] OR 2019 novel coronavirus[Title/Abstract] OR covid19[Title/Abstract] OR sars-cov-2[Title/Abstract] OR 2019-ncov[Title/Abstract] OR coronavirus disease 2019[Title/Abstract] OR coronavirus disease-19[Title/Abstract] OR Wuhan virus*[Title/Abstract] OR severe acute respiratory syndrome coronavirus 2[Title/Abstract] OR wuhan coronavirus[Title/Abstract] OR sars2[Title/Abstract] OR wuhan seafood market pneumonia virus*[Title/Abstract])) AND (("2019/12/01"[Date - Publication] : "2021/09/08"[Date - Publication]))	167,289
#2	("Coronavirus"[Mesh]) AND (("2019/12/01"[Date - Publication]: "2021/09/08"[Date - Publication]))	83,604
#3	#1 OR #2	171,191
#4	((laboratory[Title/Abstract] OR biomarker*[Title/Abstract] OR marker*[Title/Abstract] OR blood[Title/Abstract] OR serum[Title/Abstract] OR myocardial[All fields] OR heart[All fields] OR cardiac[All fields] OR cardio*[All fields] OR troponin[All fields] OR cTnI[All fields] OR myoglobin[All fields] OR Mb[All fields])) AND (("2019/12/01"[Date - Publication] : "2021/09/08"[Date - Publication]))	653,900

#5	#3 AND #4	34,606
#6	5# Filters: English	33,752

Search strategy for Web of Science Core Collection (September 8, 2021)

#	Searches	Results
#1	TS=((((((((((covid-19) OR (2019 novel coronavirus)) OR (covid19)) OR (sars-cov-2)) OR (2019-ncov)) OR (coronavirus disease 2019)) OR (coronavirus disease-19)) OR (Wuhan virus*)) OR (severe acute respiratory syndrome coronavirus 2)) OR (wuhan coronavirus)) OR (sars2)) OR (wuhan seafood market pneumonia virus*)) *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years*	184,871
#2	TS=((((laboratory OR biomarker*) OR marker*) OR blood) OR serum) *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years*	4,882,468
#3	All=(((((((myocardial OR heart) OR cardiac) OR cardio*) OR troponin) OR cTnI) OR myoglobin) OR Mb) *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years*	3,993,968
#4	#3 OR #2 *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years*	8,052,433
#5	#4 AND #1 *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years*	33,412
#6	(#4 AND #1) AND LANGUAGE: (English) *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=2019-2021*	32,258

Search strategy for Embase (September 8, 2021)

#	Searches	Results
#1	laboratory:ti,ab,kw OR biomarker*:ti,ab,kw OR marker*:ti,ab,kw OR blood:ti,ab,kw OR serum:ti,ab,kw	5,438,834
#2	myocardial OR heart OR cardiac OR cardio* OR troponin OR ctni OR myoglobin OR mb	4,611,584
#3	'covid 19':ti,ab,kw OR '2019 novel coronavirus':ti,ab,kw OR covid19:ti,ab,kw OR 'sars cov 2':ti,ab,kw OR '2019 ncov':ti,ab,kw OR 'coronavirus disease 2019':ti,ab,kw OR 'coronavirus disease-19':ti,ab,kw OR 'wuhan virus*':ti,ab,kw OR 'severe acute respiratory syndrome coronavirus 2':ti,ab,kw OR 'wuhan coronavirus':ti,ab,kw OR sars2:ti,ab,kw OR 'wuhan seafood market pneumonia virus*':ti,ab,kw	171,120
#4	#1 OR #2	8,888,780
#5	#3 AND #4	42,185
#6	#3 AND #4 AND [english]/lim AND [1-12-2019]/sd NOT [8-9-2021]/sd	40,915

Methods S3. Quality assessment

Quality Assessment Forms recommended by Agency for Healthcare Research and Quality (AHRQ) for cross-sectional study

1) Define the source of information (survey, record review)
2) List inclusion and exclusion criteria for exposed and unexposed subjects (cases and controls) or refer to previous publications
3) Indicate time period used for identifying patients
4) Indicate whether or not subjects were consecutive if not population-based
5) Indicate if evaluators of subjective components of study were masked to other aspects of the status of the participants
6) Describe any assessments undertaken for quality assurance purposes (eg.

test/retest of primary outcome measurements)

7) Explain any patient exclusions from analysis
8) Describe how confounding was assessed and/or controlled.
9) If applicable, explain how missing data were handled in the analysis
10) Summarize patient response rates and completeness of data collection
11) Clarify what follow-up, if any, was expected and the percentage of patients for which incomplete data or follow-up was obtained

The methodological quality of the studies included was assessed using an 11-item checklist which was recommended by Agency for Healthcare Research and Quality (AHRQ). An item would be scored '0' if it was answered 'NO' or 'UNCLEAR'; if it was answered 'YES', then the item scored '1'. Article quality was assessed as follows: low quality = 0-3; moderate quality = 4-7; high quality = 8-11.

Table S1. Quality assessment form

Study	A	B	C	D	E	F	G	H	I	J	K	Quality score
Arcari L et al.	1	1	1	1	1	1	1	0	1	1	1	10
Bardají´ A et al.	1	1	1	1	0	0	1	1	1	0	1	8
Bhatla A et al.	1	0	1	1	1	0	1	1	1	1	1	9
Cai Q et al.	1	1	1	1	1	1	1	1	1	1	1	11
Calvo-Fernández A et al.	1	1	1	1	0	1	1	1	1	1	0	9
Cao J et al.	1	0	1	1	1	0	1	1	1	1	1	9
Cao J et al.	1	1	1	1	1	1	1	1	1	1	1	11
Cao M et al.	1	0	1	1	1	0	1	0	1	1	1	8
Chen N et al.	1	1	1	1	1	1	1	1	1	1	1	11
Chorin E et al.	1	1	0	1	1	1	1	1	1	1	1	10
Cipriani A et al.	1	1	1	1	1	1	1	1	1	1	0	10
Deng Q et al.	1	0	1	1	1	0	1	0	1	1	0	7
Elhadi M et al.	1	1	1	1	1	1	1	1	1	1	0	10

Feng Y et al.	1	1	1	1	1	1	1	0	1	1	1	10
Ferguson J et al.	1	1	1	1	1	1	1	1	1	1	1	11
Ferrante G et al.	1	1	1	1	1	1	1	1	1	1	1	11
Franks C et al.	1	1	1	1	1	1	1	1	1	1	1	11
García de Guadiana-Romualdo L et al.	1	1	1	1	0	1	1	1	1	1	0	9
Garibaldi BT et al.	1	1	1	1	1	1	1	1	1	1	1	11
Guo T et al.	1	1	1	1	1	1	1	1	1	1	1	11
Han H et al.	1	0	1	1	1	0	1	1	1	1	0	8
Harmouch F et al.	1	1	1	1	1	1	1	1	1	1	1	11
He F et al.	1	1	1	1	1	1	1	1	1	1	1	11
He X et al.	1	1	1	1	1	1	1	1	1	1	0	10
Hu L et al.	1	1	1	1	1	1	1	1	1	1	1	11
Huang C et al.	1	1	1	1	1	1	1	1	1	1	1	11
Huang J et al.	1	1	1	1	0	0	1	1	1	0	1	8
Huang R et al.	1	1	1	1	1	1	1	1	1	1	1	11
Karbalai Saleh S et al.	1	0	1	1	1	0	1	1	1	1	1	9
Lala A et al.	1	1	1	1	1	1	1	0	1	1	1	10
Li C et al.	1	1	1	1	1	0	1	1	1	1	1	10
Li X et al.	1	0	1	1	1	0	1	1	1	1	1	9
Maeda T et al.	1	1	1	1	0	1	1	1	1	1	0	9
Majure D et al.	1	0	1	1	1	0	1	0	1	1	0	7
Manocha KK et al.	1	1	1	1	1	1	1	1	1	1	0	10
Merugu GP et al.	1	1	1	1	0	0	1	1	1	0	1	8
Mikami T et al.	1	1	1	1	1	1	1	0	1	1	1	10
Özyılmaz S et al.	1	1	1	1	0	0	1	1	1	0	1	8
Palaiodimos L et al.	1	1	1	1	1	1	1	1	1	1	1	11

Peiró ÓM et al.	1	1	1	1	1	1	1	1	1	1	0	10
Price-Haywood E et al.	1	1	1	1	1	1	1	1	1	1	1	11
Qin J et al.	1	1	1	1	1	1	1	1	1	1	1	11
Richardson S et al.	1	1	1	1	1	1	1	1	1	1	1	11
Schiavone M et al.	1	1	1	1	1	0	1	1	1	1	1	10
Shah P et al.	1	1	1	1	0	0	1	1	1	0	1	8
Shen Y et al.	1	0	1	1	1	1	1	1	1	1	1	10
Singh N et al.	1	1	1	1	1	1	1	1	1	1	1	11
Stefanini G et al.	1	0	1	1	1	0	1	1	1	1	1	9
Suleyman G et al.	1	1	1	1	1	1	1	0	1	1	1	10
Tanboğa IH et al.	1	1	1	1	0	1	1	1	1	1	0	9
Tomasoni D et al.	1	1	1	1	1	1	1	1	1	1	0	10
Wang D et al.	1	0	1	1	1	0	1	1	1	1	0	8
Wang Z et al.	1	0	1	1	1	0	1	0	1	1	1	8
Wei J et al.	1	1	1	1	1	1	1	1	1	1	1	11
Wu Y et al.	1	1	1	1	1	1	1	1	1	1	1	11
Xu P et al.	1	0	1	1	1	0	1	1	1	1	1	9
Zeng J et al.	1	1	0	1	1	1	1	0	1	1	1	9
Zhang G et al.	1	0	1	1	1	1	1	0	1	1	0	8
Zhang Q et al.	1	1	1	1	1	1	1	1	1	1	0	10
Zhang Y et al.	1	0	1	1	1	1	1	1	1	1	1	10
Zhao M et al.	1	0	1	1	1	0	1	0	1	1	1	8
Zhao X et al.	1	0	1	1	1	1	1	0	1	1	1	9
Zhou F et al.	1	0	1	1	1	1	1	1	1	1	1	10

Table S2 Characteristics of the included studies

Author	Study period	Country (city)	No.	Median age (IQR)	Male n (%)	CSR n (%)	CFR n (%)	Cardiovascular condition	Mb	cTnI	Outcome	Quality Score [a]
Arcari L et al.[3]	2020.03.15 - 2020.04.30	Rome, Italy	111	72.0 (17.0) [b]	51 (46.0)	NA	23 (21.0)	CAD, 12 (11.0); HF, 8 (7.0)	NA	Average level of cTnI,17 (5-47) pg/mL; cut-off value, 14 pg/ml; elevated patients, 39/103 (37.9%)	Death	10
Bardajı´ A et al.[4]	2020.03.16-2020.04.16	Spain	186	67.5 (52.5-77.5)	111 (59.7)	NA	37 (19.9)	CAD, 20 (10.8); HF, 14 (7.5)	NA	Elevated patients, 41 (22.0%)	Death, admission to ICU	8
Bhatla A et al.[5]	2020.03.06 - 2020.03.19	Pennsylvania, the U.S.	700	50.0 (18.0) [b]	314 (45.0)	NA	30 (4.3)	CAD, 76 (11.0); HF, 88 (13.0); BNP, 2940 (7962) pg/mL	NA	Cut-off value, 0.01 ng/mL; elevated patients, 82/373 (22.0%)	NA	9
Cai Q et al.[6]	2020.01.11 - 2020.03.06	Shenzhen, China	298	47.5 (33.0-61.0)	145 (48.66)	58 (19.5)	3 (1.0)	CAD, 25 (8.4); HF, 7 (2.3)	Average level of Mb, 37.1 (29.2-51.5) ug/L; elevated patients, 10/260 (3.8%)	NA	Death, discharge	11
Calvo-Fernández A et al.[7]	2020.02.27-2020.05.01	Barcelona, Spain	872	62.3 (18.1) [b]	486 (55.7)	NA	104 (11.9)	CAD, 59 (6.83); HF, 41 (4.73)	NA	Cut-off value, 14.0 ng/L; elevated patients, 225/651 (34.6%)	Death, admission to ICU mechanical ventilation	9
Cao J et al.[8]	2020.01.03 - 2020.02.15	Wuhan, China	102	54.0 (37.0-67.0)	53 (52.0)	NA	17 (16.7)	CAD, 5 (4.9); BNP, 12.2 (0-63.1) pg/mL, NT-pro BNP, 417 (132-1800) pg/mL	NA	Average level of cTnI,8.0 (3.0-35.7) pg/mL; cut-off value, 0.026 ng/mL； elevated patients, 15/55 (27.3%)	Discharge, death	9
Cao J et al.[9]	2020.02.06 - 2020.02.21	Wuhan, China	244	62.6 (13.4) [b]	133 (54.5)	153 (62.7)	14 (5.74)	NA	Average level of Mb in severe patients, 39.35 (29.21-74.19) ug/L; Cut-off value, 110 ug/L	Cut-off value, 0.04 ng/mL； elevated patients, 27/244 (11.1%)	Severe COVID-19, death, mechanic ventilation	11
Cao M et al.[10]	2020.01.20 - 2020.02.15	Shanghai, China	198	50.1 (16.3) [b]	101 (51.0)	19 (9.6)	NA	CAD, 12 (6.0)	Average level of Mb,5.9 (2.8-15.7) ug/L; cut-off value, 48.8 ug/L； elevated patients, 33/194 (17.0%)	Average level of cTnI,0.02 (0.01-0.04) ng/ml; cut-off value, 0.04 ng/mL； elevated patients, 22/194 (11.3%)	Severe COVID-19	8
Chen N et al.[11]	2020.01.01 - 2020.01.20	Wuhan, China	99	55.0 (13.1) [b]	67 (68.0)	NA	11 (11.0)	CAD, 40 (40.0)	Average level of Mb,49.5 (32.2-99.8) ug/L; cut-off value, 146.9 ug/L; elevated patients, 15 (15.2%)	NA	Discharge, death	11
Chorin E et al.[12]	NA	New York, the U.S.	204	64.0 (13.0) [b]	156 (76.0)	NA	50 (23.0)	CAD, 25 (12.0); HF, 7 (3.0)	NA	Average level of cTnI, 0.02 (0.01-0.04) ng/Ml; cut-off value, 0.05 ng/mL; elevated patients, 84 (41.2%)	Death	10
Cipriani A et al.[13]	2020.02.26 - 2020.03.31	Italy	109	71.0 (60.0-81.0)	73 (67.0)	NA	20 (18.3)	CAD, 18 (17.0); HF, 16 (15.0%); BNP, 90 (22-262) pg/ml	NA	Average level of cTnI,18.0 (7.0-96.0) ng/L; cut-off value, 32 ng/L for males,16 ng/L for females; elevated patients, 46 (42.2%)	Death, admission to ICU, discharge	10
Deng Q et al.[14]	2020.01.06 - 2020.02.20	Wuhan, China	112	65.0 (49.0-70.8)	57 (59.0)	67 (59.8)	14 (12.5)	CAD, 15 (13.4); HF, 6 (5.4); NT-pro BNP, 430.1 (100.6-2859.3) ng/L	NA	Average level of cTnI, 0.01 (0.00-0.14) ng/ml; cut-off value, 0.04 ng/mL； elevated patients, 42 (37.5%)	Severe COVID-19, death	7

Elhadi M et al.[15]	2020.03.24-2020.12.03	Libya	1207	56 (40-64)	811 (67.2)	NA	149 (12.3)	CAD, 25 (2.1)	NA	Cut-off value, 26 pg/mL; elevated patients, 90/292 (30.8%)	Death, admission to ICU	10
Feng Y et al.[16]	2020.01.01 - 2020.02.25	3 provinces, China	476	53.0 (40.0-64.0)	271 (56.9)	124 (26.5)	38 (8.0)	CAD, 38 (8.0); BNP, 40.85 (21.64-79.37) pg/ml	Average level of Mb,18.85 (4.8-51.48) ug/L	Elevated patients, 86/384 (22.4%)	Death, discharge, severe COVID-19	10
Ferguson J et al.[17]	2020.03.13 - 2020.05.02	Standford, the U.S.	72	NA	NA	NA	6 (8.3)	NA	NA	Cut-off value, 0.055 ng/mL; elevated patients, 2/45 (4.4%)	Death, mechanical ventilation, admission to ICU	11
Ferrante G et al.[18]	2020.02.25 - 2020.04.02	Helsinki, Finland	332	66.9 (55.4-75.5)	237 (71.4)	NA	68 (20.5)	CAD, 49 (14.5); BNP, 72.5 (34.5-198.0) pg/mL	NA	Average level of cTnI, 11.4 (4.7-37.3) mg/L; cut-off value, 0.02 ng/L; elevated patients, 123 (37.0%)	Death, admission to ICU	11
Franks C et al.[19]	NA	Washington, the U.S.	182	64.0 (19.0-98.0)	103 (56.6)	NA	34 (18.7)	NA	NA	Cut-off value, 0.03 ng/mL; elevated patients, 80/143 (55.9%)	Death	11
García de Guadiana-Romualdo L et al.[20]	2020.03.01-2020.05.12	Spain	1280	67 (55-77)	750 (58.6)	NA	187 (14.6)	CAD, 328 (25.6)	NA	Elevated patients, 344 (26.9%)	Death, admission to ICU	9
Garibaldi BT et al.[21]	2020.03.04-2020.06.27	Washington, the U.S.	832	63 (49-75)	443 (53.2)	171 (20.6)	131 (15.7)	CAD, 266 (32.0); HF, 127 (15.0); NT-pro BNP 214 (45-960) pg/mL	NA	Elevated patients, 194/682 (28.4%)	Death, severe COVID-19	11
Guo T et al.[22]	2020.01.23 - 2020.02.25	Wuhan, China	187	58.5 (14.7) [b]	91 (48.7)	NA	43 (23.0)	CAD, 21 (11.2); NT-pro BNP, 268.4 (75.3-689.1) pg/mL	Average level of Mb,38.5 (21.0-78.0) ug/L	Elevated patients, 52 (27.8%)	Death	11
Han H et al.[23]	2020.01.01 - 2020.02.18	Wuhan, China	273	NA	97 (35.5)	75 (27.5)	24 (8.79)	NA	Cut-off value, 110 ug/L; elevated patients, 29/273 (10.6%)	Cut-off value, 0.04 ng/mL; elevated patients, 27/273 (9.9%)	Death, severe COVID-19	8
Harmouch F et al.[24]	2020.03.01 - 2020.04.15	Pennsylvania, the U.S.	560	63.0 (24.0) [b]	319 (57.1)	NA	81 (14.4)	Vascular disease, 36 (6.4); HF, 54 (9.6)	NA	Cut-off value, 0.05 ng/mL; elevated patients, 97/482 (20.1%)	Death, mechanical ventilation, admission to ICU	11
He F et al.[25]	2020.01.15 - 2020.03.10	Guangzhou, China	288	48.5 (34.3-62.0)	131 (45.5)	30 (10.4)	1 (0.3)	CAD, 85 (29.5); BNP, 35 (13-117.5) pg/mL	Elevated patients, 8/276 (2.9%)	Cut-off value, 0.03 ng/mL; elevated patients, 22/190 (11.6%);	Death, admission to ICU	11
He X et al.[26]	2020.02.01-2020.03.29	Wuhan, China	1031	63 (52-70)	538 (52.2)	501 (48.6)	165 (16.0)	CAD, 83 (8.1); NT-pro BNP 124 (43-374) pg/mL	NA	Average level of cTnI, 5.3 (2.5-14.0) pg/Ml; elevated patients, 215 (20.9%)	Death	10
Hu L et al.[27]	2020.01.08 - 2020.03.10	Wuhan, China	323	61 (23-91) [c]	166 (51.4)	172 (53.3)	35 (10.8)	CAD, 41 (12.7)	NA	Cut-off value, 0.04 pg/mL; elevated patients, 68 (21.1%)	Death, severe COVID-19, mechanical ventilation	11
Huang C et al.[28]	2019.12 - 2020.01.02	Wuhan, China	41	49.0 (41.0-58.0)	30 (73.0)	NA	6 (15.0)	CAD, 6 (15.0)	NA	Average level of cTnI,3.4 (1.1-9.1) pg/mL; cut-off value, 0.028 ng/mL； elevated patients, 5/41 (12.2%)	Death, severe COVID-19, discharge	11
Huang J et al.[29]	2020.01.17-2020.02.16	Guangzhou, China	98	44.0 (33.0-62.3)	46 (47.0)	22 (22.4)	1 (1.0)	CAD, 6 (6.0); BNP 119 (54-392) pg/mL	NA	Cut-off value, 0.0229 ng/Ml; elevated patients, 7 (7.1%)	Death, discharge, severe COVID-19	8

Huang R et al.[30]	2020.01.22 - 2020.02.10	Jiangsu, China	202	44.0 (33.0-54.0)	116 (57.4)	23 (11.4)	0	CAD, 5 (2.5)	NA	Elevated patients, 2/103 (1.9%)	Admission to ICU, mechanical ventilation, severe COVID-19	11
Karbalai Saleh S et al.[31]	2020.03 - 2020.05	South Tehran, Iran	386	59.5 (15.8) [b]	236 (61.1)	NA	77 (19.9)	CAD, 97 (25.1)	NA	cut-off value, 26 ng/L for males,11 ng/L for females; elevated patients, 115 (29.8%)	Death, admission to ICU	9
Lala A et al.[32]	2020.02.27 - 2020.04.12	New York, the U.S.	2736	66.4 (15.8)	1630 (59.6)	NA	506 (18.5)	CAD, 453 (16.6); HF, 276 (10.1)	NA	Cut-off value, 0.03 ng/mL; OR for in-hospital mortality,1.75 (1.37-2.24); elevated patients, 985 (36.0%)	Death	10
Li C et al.[33]	2020.01.29 - 2020.04.01	Wuhan, China	2068	63.0 (51.0-70.0)	1005 (48.6)	476 (23.0)	183 (8.8)	CAD, 182 (8.8); HF, 14 (0.7); NT-pro BNP 108 (36-370) pg/mL	Average level of Mb,40.7 (28.4-73.8) ug/L; elevated patients, 174/1554 (11.2%)	Average level of cTnI,4.2 (1.9-11.0) pg/mL; elevated patients, 181 (8.8%)	Death, severe COVID-19	10
Li X et al.[34]	2020.01.26 - 2020.03.03	Wuhan, China	548	60.0 (48.0-69.0)	279 (50.9)	269 (49.1)	90/545 (16.5)	CAD, 34 (6.2)	NA	Cut-off value, 15.6 pg/mL; elevated patients, 119 (21.7%)	Discharge, death, severe COVID-19	9
Maeda T et al.[35]	2020.03.13-2020.03.31	New York, the U.S.	181	64.0 (16.6) [b]	101 (55.8)	NA	38/178 (21.3)	CAD, 36 (19.9); HF, 24/180 (13.3)	NA	Elevated patients, 54 (29.8%)	Death	9
Majure D et al.[36]	2020.03.01 - 2020.04.27	New York, the U.S.	6247	66.0 (56.0-77.0)	3740 (60.0)	NA	1397 (22.0)	CAD, 833 (13.0); HF, 529 (9.0)	NA	Cut-off value, 0.045 ng/mL; elevated patients, 1821 (29.1%)	Death, admission to ICU, mechanical ventilation	7
Manocha KK et al.[37]	2020.03.03-2020.04.06	New York, the U.S.	446	65.0 (15.2) [b]	291 (65.3)	NA	95 (21.3)	CAD, 94 (21.1); HF, 38 (8.5) BNP 84 (25-300) pg/mL	NA	Average level of cTnI, 0.05 (0-0.34) ng/Ml; cut-off value, 0.34 ng/mL; elevated patients, 112 (25.1%)	Death, admission to ICU	10
Merugu GP et al. [38]	2020.03.25-2020.06.16	Northwest Ohio, the U.S.	217	63.1 (17.8) [b]	102 (47.0)	NA	23 (10.6)	NA	NA	Elevated patients, 34/201 (16.9%)	Death	8
Mikami T et al.[39]	2020.03.12 - 2020.04.17	New York, the U.S.	6493	59.0 (43.0-72.0)	3538 (54.5)	NA	858 (13.2)	NA	NA	Average level of cTnI,0.03 (0.02-0.10) ng/dl; cut-off value, 0.03 ng/dL; elevated patients,1312/2526 (51.9%)	Death	10
Özyılmaz S et al.[40]	2020.03.20-2020.06.20	Istanbul, Turkey	105	45 (20-87) [c]	76 (72.4)	NA	9 (8.6)	CAD, 14 (21.1)	NA	Average level of cTnI, 2.6 (0-1774.5) pg/mL [c]; cut-off value, 7.8 ng/mL elevated patients, 21 (20.0%)	Death	8
Palaiodimos L et al.[41]	2020.03.09 - 2020.04.12	New York, the U.S.	200	64.0 (50.0-73.5)	98 (49.0)	NA	48 (24.0)	CAD, 33 (16.5); HF, 34 (17.0)	NA	Cut-off value, 0.01 ng/mL; elevated patients, 56 (28.0%)	Mortality, intubation, O2 requirement, ARDS, ICU, AKI, RRT, length of stay	11
Peiró ÓM et al.[42]	2020.03.16-2020.05.15	Tarragona, Spain	196	67.5 (53.5-78.0)	117 (59.7)	NA	37 (18.9)	CAD, 19 (9.7); HF, 14 (7.1)	NA	Average level of cTnI, 14 (4-37) ng/L; cut-off value, 21 ng/L elevated patients, 77 (39.3%)	Death, admission to ICU mechanical ventilation	10
Price-Haywood E et al.[43]	2020.03.01 - 2020.04.11	Louisiana, the U.S.	3481	54.0	1394 (40.0)	NA	326/1382 (23.6)	CAD, 139 (4.0); HF, 128 (3.7)	NA	Cut-off value, 0.06 ng/mL; elevated patients, 270/1084 (24.9%)	Death, admission to ICU	11

Qin J et al. [44]	2019.12.31 - 2020.03.04	Hubei, China	3219	57.0 (45.0-66.0)	1535 (47.7)	NA	193 (6.0)	CAD, 206 (6.4)	Elevated patients, 228/1895 (12.0%); HR for in-hospital mortality,6.84 (4.95-9.45) AUC for mortality,0.83 (0.80-0.86)	Elevated patients, 95/1462 (6.5%); HR for in-hospital mortality,9.59 (6.36-14.47); AUC for in-hospital mortality,0.78 (0.73-0.84)	Death	11
Richardson S et al.[45]	2020.03.01 - 2020.04.04	New York, the U.S.	5700	63.0 (52.0-75.0)	3437 (60.3)	NA	553/2634 (21.0)	CAD, 595 (11.1); HF, 371 (6.9); BNP, 385.5 (160-1996.8), n=1818	NA	Elevated patients,801/3533 (22.7%)	Admission to ICU, mechanical ventilation, kidney replacement therapy, Death	11
Schiavone M et al.[46]	2020.02.23 - 2020.05.01	Italy	674	60.8 (15.9) [b]	406 (60.2)	NA	105 (15.6)	HF, 111 (16.5)	NA	Average level of cTnI,18 (8-40) ng/L; elevated patients, 130 (19.3%)	Death, admission to ICU, mechanical ventilation	10
Shah P et al.[47]	2020.03.02-2020.06.07	Georgia, the U.S.	309	NA	178 (57.3)	NA	66 (21.4)	CAD, 28 (9.1); HF, 65 (21.0)	NA	Elevated patients, 116 (37.5%)	Death, admission to ICU mechanical ventilation	8
Shen Y et al.[48]	2020.01.20 - 2020.02.29	Shanghai, China	325	51.0 (31.0-64.0)	168 (51.7)	26 (8.0)	3 (0.92)	NA	Cut-off value, 48.8 ug/L; elevated patients, 28/325 (8.6%)	Cut-off value, 0.04 ng/mL; elevated patients, 80/325 (24.6%)	Death, discharge	10
Singh N et al.[49]	2020.03.23 - 2020.04.13	Chicago, the U.S.	276	62.0 (50.0-73.0)	130 (47.1)	NA	26 (9.4)	Vascular disease, 49 (17.8); HF, 56 (20.3)	NA	Cut-off value, 0.017 ng/mL; elevated patients, 132/276 (47.8%) OR for in-hospital mortality, 4.43 (1.61-12.19)	Death	11
Stefanini G et al.[50]	2019.12 - 2020.04.01	Italy	397	66.7 (55.0-75.7)	267 (67.3)	NA	92 (23.2)	Prior MI, 33/395 (8.4); HF, 18/395 (4.6); BNP, 67 (30-191) pg/mL	NA	Average level of cTnI, max 10.8 (4.3-39.5) ng/L, baseline 7.8 (4.5-25.6) ng/L; elevated patients, 130 (32.7%)	Death, admission to ICU, discharge	9
Suleyman G et al.[51]	2020.03.09 - 2020.03.27	Michigan, the U.S.	463	57.5 (16.8)	204 (44.1)	NA	16 (3.5)	CAD, 59 (12.7); HF, 49 (10.6)	NA	Elevated patients, 107 (23.1%)	Death, admmission to ICU	10
Tanboğa IH et al.[52]	2020.03.11-2020.06.22	Turkey	14855	49 (36-62)	8272 (54.0)	NA	882 (5.9)	CAD, 2341 (15.3); HF, 776 (5.1)	NA	Average level of cTnI 0.08 (0.00-0.28) ng/mL; elevated patients, 1027 (6.9%)	Death, admission to ICU mechanical ventilation	9
Tomasoni D et al.[53]	2020.03.01-2020.04.09	Italy	692	67.4 (13.2) [b]	481 (69.5)	NA	163 (23.6)	CAD, 148 (21.4) HF, 90 (13.0) NT-pro BNP 303 (96-1201) pg/mL	NA	Elevated patients, 272/605 (45.0%)	Death	10
Wang D et al.[54]	2020.01.01 - 2020.02.03	Wuhan, China	138	56.0 (42.0-68.0)	75 (54.3)	NA	6 (4.3)	CAD, 20 (14.5)	NA	Average level of cTnI,6.4 (2.8-18.5) pg/mL; cut-off value, 0.0262 ng/mL; Elevated patients, 10 (7.2%)	Admission to ICU	8
Wang Z et al.[55]	2019.12.30 - 2020.02.29	Wuhan, China	293	59.2 (42.8-73.1)	138 (47.1)	NA	NA	CAD, 21 (7.2)	Average level of Mb,57.6 (30.8-116.4) ug/L; cut-off value, 110 ug/L； elevated patients, 58/213 (27.2%)	Average level of cTnI,0.007 (0.006-0.046) ng/mL; cut-off value, 0.0796 ng/mL； elevated patients, 36/216 (16.7%)	Death	8
Wei J et al.[56]	2020.01.16 - 2020.03.10	Sichuan, China	101	49.0 (34.0-62.0)	54 (53.5)	37 (36.6)	3 (3.0)	CAD, 5 (5.0); NT-pro BNP, 71.2 (31.6-237.5) pg/mL	NA	Average level of cTnI,6.8 (4.3-10.1) pg/mL; cut-off value, 0.014 ng/mL； elevated patients, 16 (15.8%)	Death, severe case, admission to ICU, mechanical ventilation	11

Wu Y et al.[57]	2020.02.10 - 2020.03.30	Wuhan, China	125	55.0 (40.0-68.5)	66 (52.8)	2 (1.6)	0	CAD, 11 (8.8); BNP, 65.0 (23.0-178.0) pg/mL	Average level of Mb,35.0 (27.7-75.65) ug/L; cut-off value, 154.9 ug/L; elevated patients, 14 (11.2%)	Average level of cTnI,3.9 (1.9-10.3) pg/ml; cut-off value, 0.0342 ng/mL; elevated patients, 10 (8.0%)	Long-term hospitalization	11
Xu P et al.[58]	2020.01.10 - 2020.03.13	8 proviences, China	703	46.1 (15.2) [b]	382 (54.3)	55 (7.8)	33 (4.7)	CAD, 35 (5.0)	Elevated patients, 33/181 (18.2%)	NA	Death, admission to ICU, mechanical ventilation	9
Zeng J et al.[59]	2020.01.11 - 2020.04.01	Shenzhen, China	416	NA	198 (47.6)	NA	3 (0.7)	CAD, 13 (3.1); HF, 5/57 (8.8)	Cut-off value, 100 ug/L; elevated patients, 30/174 (17.2%)	Cut-off value, 0.026 ng/mL; elevated patients, 29/345 (8.4%)	Death, discharge	9
Zhang G et al.[60]	2020.01.02 - 2020.02.10	Wuhan, China	221	55.0 (39.0-66.5)	108 (48.9)	55	12 (5.4)	CAD, 22 (10.0)	NA	Average level of cTnI,7.6 (3.6-21.5) pg/mL; cut-off value, 0.0262 ng/mL; elevated patients, 17 (7.7%)	Discharge, death, severe COVID-19	8
Zhang Q et al.[61]	2020.01.23-2020.02.25	Jilin, China	41	45 (31-53)	23 (56.1)	8 (19.5)	NA	CAD, 1 (2.4)	Average level of Mb, 26.0 (19.7-118.6) ug/L elevated patients, 11 (26.8%)	Average level of cTnI; 1.5 (0.8-5.0) ng/mL; elevated patients, 41 (100%)	Severe COVID-19	10
Zhang Y et al.[62]	2020.02.28 - 2020.04.04	Wuhan, China	166	62.7 (14.2) [b]	85 (51.2)	136 (81.9)	24 (14.5)	CAD, 30 (18.1); NT-proBNP, 179.0 (67.0-457.0) pg/mL	Average level of Mb,54.8 (33.8-127.2) ug/L; cut-off value, 106 ug/L; elevated patients, 28/166 (16.9%)	Average level of cTnI,5.0 (2.2-10.7) pg/mL; cut-off value, 0.0156 ng/mL; elevated patients, 17 /166 (10.2%)	Discharge, death	10
Zhao M et al.[63]	2020.01.01 - 2020.02.14	Wuhan, China	1000	61.0 (46.0-70.0)	466 (46.6)	615 (61.5)	119 (11.9)	CAD, 60 (6.0)	Average level of Mb, 44.54 (28.5-85.05) ug/L; cut-off value, 110 ug/L; elevated patients, 132/754 (17.5%)	Average level of cTnI,0.006 (0.006-0.018) ng/mL; cut-off value, 0.0796 ng/mL; elevated patients, 66/758 (8.7%)	Death, discharge	8
Zhao X et al.[64]	2020.01.16 - 2020.02.10	Hubei, China	91	46.0	49 (53.8)	30 (33.0)	2 (2.2)	HF, 14 (15.4)	NA	Cut-off value, 0.01 ng/mL; elevated patients, 3/88 (3.4%)	Death, discharge	9
Zhou F et al.[65]	2019.12.29 - 2020.01.31	Wuhan, China	191	56.0 (46.0-67.0)	119 (62.0)	66 (34.6)	54 (28.3)	CAD, 15 (8.0); HF, 44 (23.0)	NA	Average level of cTnI,4.1 (2.0-14.1) ng/mL; cut-off value, 28 ng/mL; elevated patients, 24/145 (16.6%)	Death, admission to ICU	10

No., number of COVID-19 confirmed patients; IQR, interquartile range; CSR, case severity rate; CFR, case fatality rate; Mb, myoglobin; cTnI, cardiac troponin I; NA, data not available.

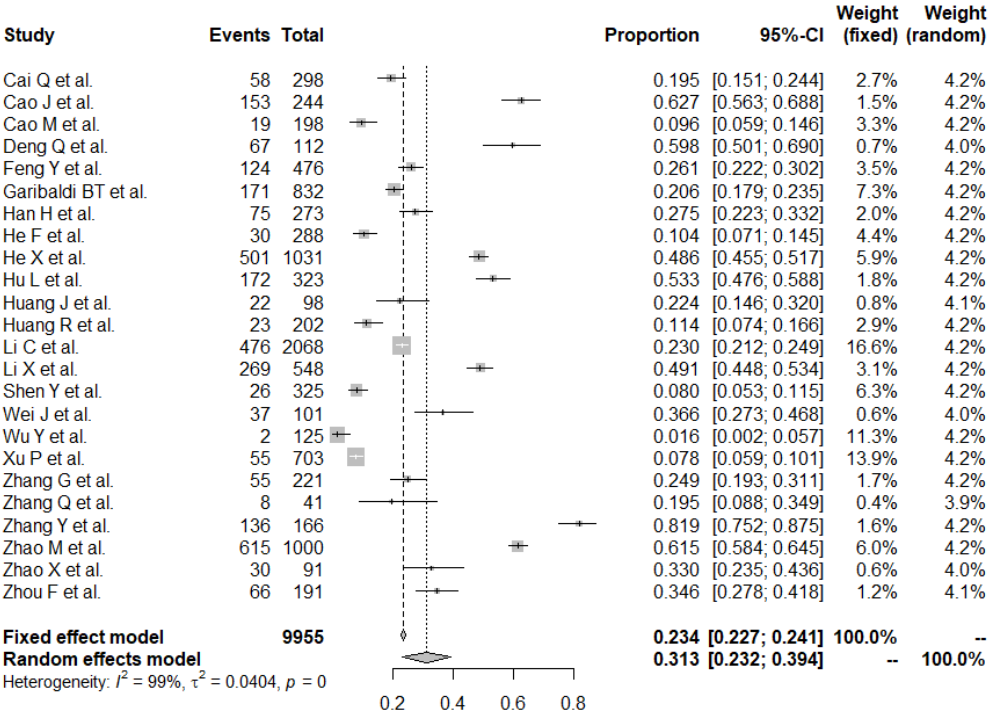
[a] Study quality was assessed using the Quality Assessment Forms recommended by Agency for Healthcare Research and Quality (AHRQ);

[b] Mean (standard deviation);

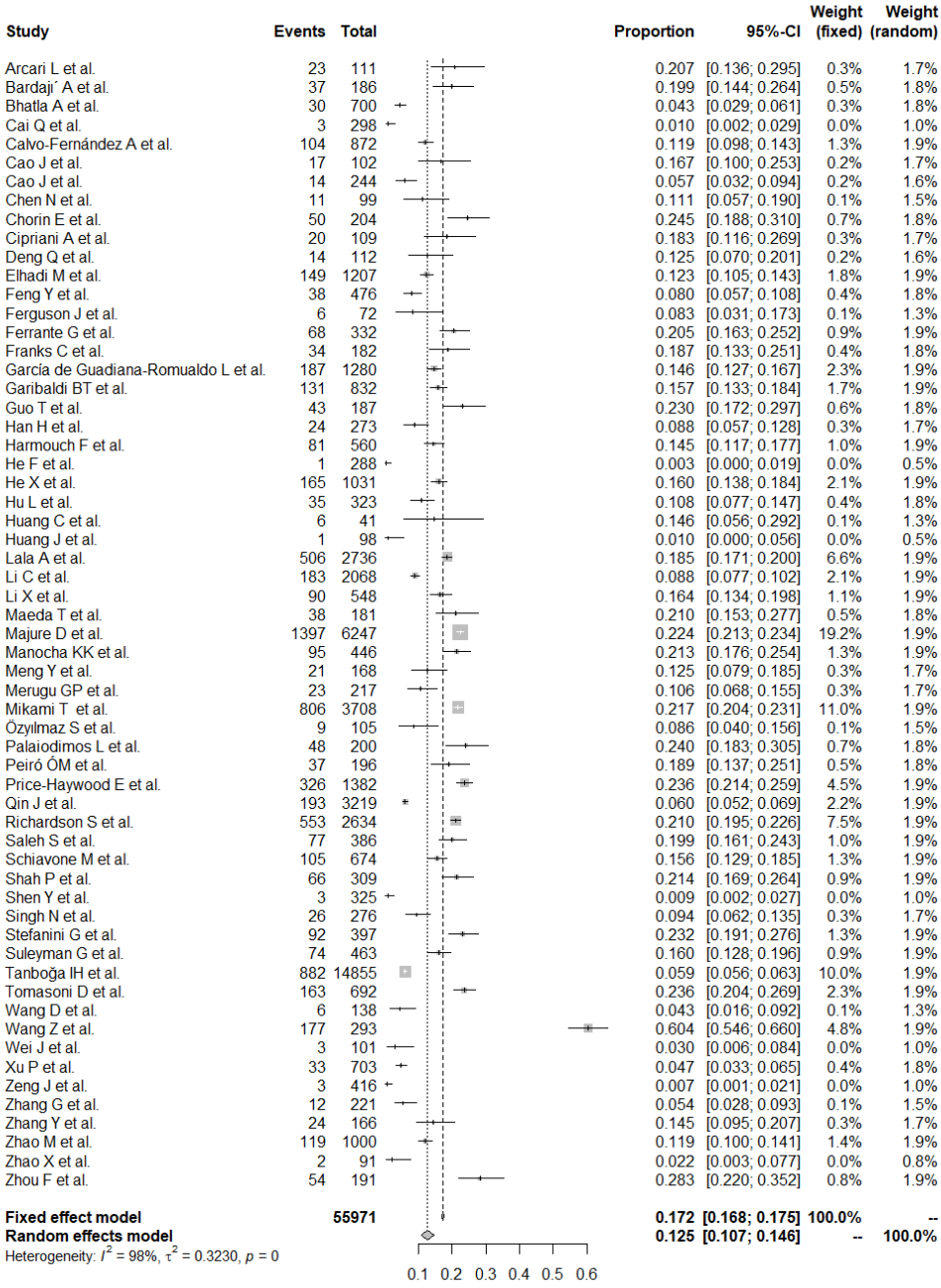
[c] Median (range).

Figure S1. Forest plot of the overall prevalence of severe illness (A) and in-hospital mortality (B) and ICU-admission (C) of COVID-19 patients

A



B


Study
Events
Total
Proportion
95%-CI
Weight (fixed)
Weight (random)
Arcari L et al. 23 111 0.207 [0.136; 0.295] 0.3% 1.7%
Bardaji´ A et al. 37 186 0.199 [0.144; 0.264] 0.5% 1.8%
Bhatla A et al. 30 700 0.043 [0.029; 0.061] 0.3% 1.8%
Cai Q et al. 3 298 0.010 [0.002; 0.029] 0.0% 1.0%
Calvo-Fernández A et al. 104 872 0.119 [0.098; 0.143] 1.3% 1.9%
Cao J et al. 17 102 0.167 [0.100; 0.253] 0.2% 1.7%
Cao J et al. 14 244 0.057 [0.032; 0.094] 0.2% 1.6%
Chen N et al. 11 99 0.111 [0.057; 0.190] 0.1% 1.5%
Chorin E et al. 50 204 0.245 [0.188; 0.310] 0.7% 1.8%
Cipriani A et al. 20 109 0.183 [0.116; 0.269] 0.3% 1.7%
Deng Q et al. 14 112 0.125 [0.070; 0.201] 0.2% 1.6%
Elhadi M et al. 149 1207 0.123 [0.105; 0.143] 1.8% 1.9%
Feng Y et al. 38 476 0.080 [0.057; 0.108] 0.4% 1.8%
Ferguson J et al. 6 72 0.083 [0.031; 0.173] 0.1% 1.3%
Ferrante G et al. 68 332 0.205 [0.163; 0.252] 0.9% 1.9%
Franks C et al. 34 182 0.187 [0.133; 0.251] 0.4% 1.8%
García de Guadiana-Romualdo L et al. 187 1280 0.146 [0.127; 0.167] 2.3% 1.9%
Garibaldi BT et al. 131 832 0.157 [0.133; 0.184] 1.7% 1.9%
Guo T et al. 43 187 0.230 [0.172; 0.297] 0.6% 1.8%
Han H et al. 24 273 0.088 [0.057; 0.128] 0.3% 1.7%
Harmouch F et al. 81 560 0.145 [0.117; 0.177] 1.0% 1.9%
He F et al. 1 288 0.003 [0.000; 0.019] 0.0% 0.5%
He X et al. 165 1031 0.160 [0.138; 0.184] 2.1% 1.9%
Hu L et al. 35 323 0.108 [0.077; 0.147] 0.4% 1.8%
Huang C et al. 6 41 0.146 [0.056; 0.292] 0.1% 1.3%
Huang J et al. 1 98 0.010 [0.000; 0.056] 0.0% 0.5%
Lala A et al. 506 2736 0.185 [0.171; 0.200] 6.6% 1.9%
Li C et al. 183 2068 0.088 [0.077; 0.102] 2.1% 1.9%
Li X et al. 90 548 0.164 [0.134; 0.198] 1.1% 1.9%
Maeda T et al. 38 181 0.210 [0.153; 0.277] 0.5% 1.8%
Majure D et al. 1397 6247 0.224 [0.213; 0.234] 19.2% 1.9%
Manocha KK et al. 95 446 0.213 [0.176; 0.254] 1.3% 1.9%
Meng Y et al. 21 168 0.125 [0.079; 0.185] 0.3% 1.7%
Merugu GP et al. 23 217 0.106 [0.068; 0.155] 0.3% 1.7%
Mikami T et al. 806 3708 0.217 [0.204; 0.231] 11.0% 1.9%
Özyılmaz S et al. 9 105 0.086 [0.040; 0.156] 0.1% 1.5%
Palaiodimos L et al. 48 200 0.240 [0.183; 0.305] 0.7% 1.8%
Peiró ÓM et al. 37 196 0.189 [0.137; 0.251] 0.5% 1.8%
Price-Haywood E et al. 326 1382 0.236 [0.214; 0.259] 4.5% 1.9%
Qin J et al. 193 3219 0.060 [0.052; 0.069] 2.2% 1.9%
Richardson S et al. 553 2634 0.210 [0.195; 0.226] 7.5% 1.9%
Saleh S et al. 77 386 0.199 [0.161; 0.243] 1.0% 1.9%
Schiavone M et al. 105 674 0.156 [0.129; 0.185] 1.3% 1.9%
Shah P et al. 66 309 0.214 [0.169; 0.264] 0.9% 1.9%
Shen Y et al. 3 325 0.009 [0.002; 0.027] 0.0% 1.0%
Singh N et al. 26 276 0.094 [0.062; 0.135] 0.3% 1.7%
Stefanini G et al. 92 397 0.232 [0.191; 0.276] 1.3% 1.9%
Suleyman G et al. 74 463 0.160 [0.128; 0.196] 0.9% 1.9%
Tanboğa IH et al. 882 14855 0.059 [0.056; 0.063] 10.0% 1.9%
Tomasoni D et al. 163 692 0.236 [0.204; 0.269] 2.3% 1.9%
Wang D et al. 6 138 0.043 [0.016; 0.092] 0.1% 1.3%
Wang Z et al. 177 293 0.604 [0.546; 0.660] 4.8% 1.9%
Wei J et al. 3 101 0.030 [0.006; 0.084] 0.0% 1.0%
Xu P et al. 33 703 0.047 [0.033; 0.065] 0.4% 1.8%
Zeng J et al. 3 416 0.007 [0.001; 0.021] 0.0% 1.0%
Zhang G et al. 12 221 0.054 [0.028; 0.093] 0.1% 1.5%
Zhang Y et al. 24 166 0.145 [0.095; 0.207] 0.3% 1.7%
Zhao M et al. 119 1000 0.119 [0.100; 0.141] 1.4% 1.9%
Zhao X et al. 2 91 0.022 [0.003; 0.077] 0.0% 0.8%
Zhou F et al. 54 191 0.283 [0.220; 0.352] 0.8% 1.9%
Fixed effect model 55971 0.172 [0.168; 0.175] 100.0% --
Random effects model 0.125 [0.107; 0.146] -- 100.0%
Heterogeneity: I^2 = 98%, τ^2 = 0.3230, p = 0
0.1 0.2 0.3 0.4 0.5 0.6

C

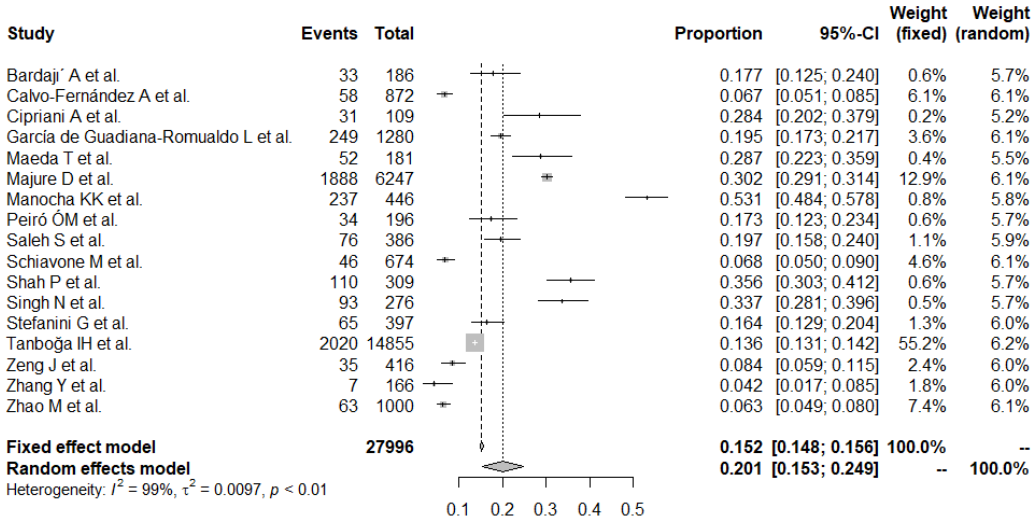
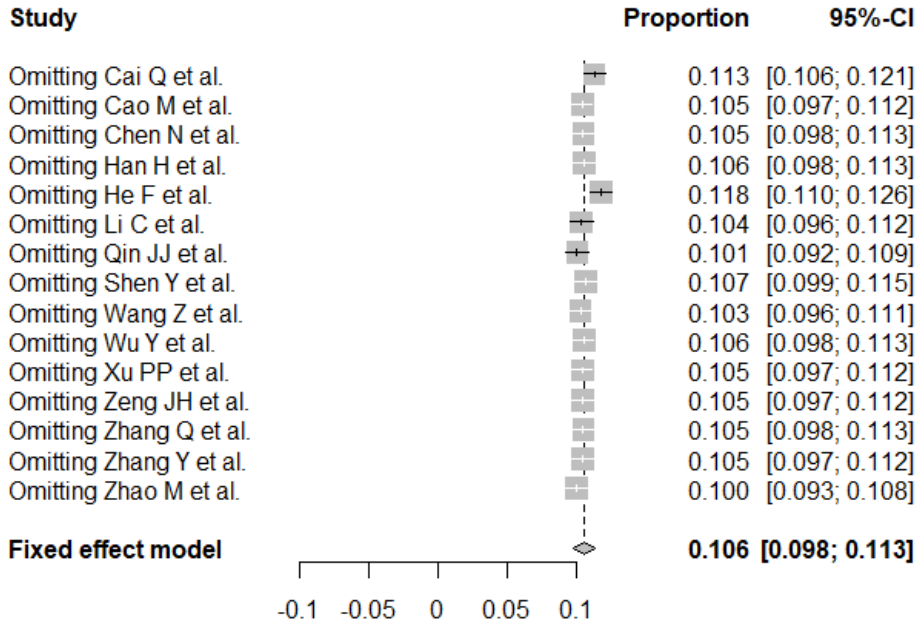
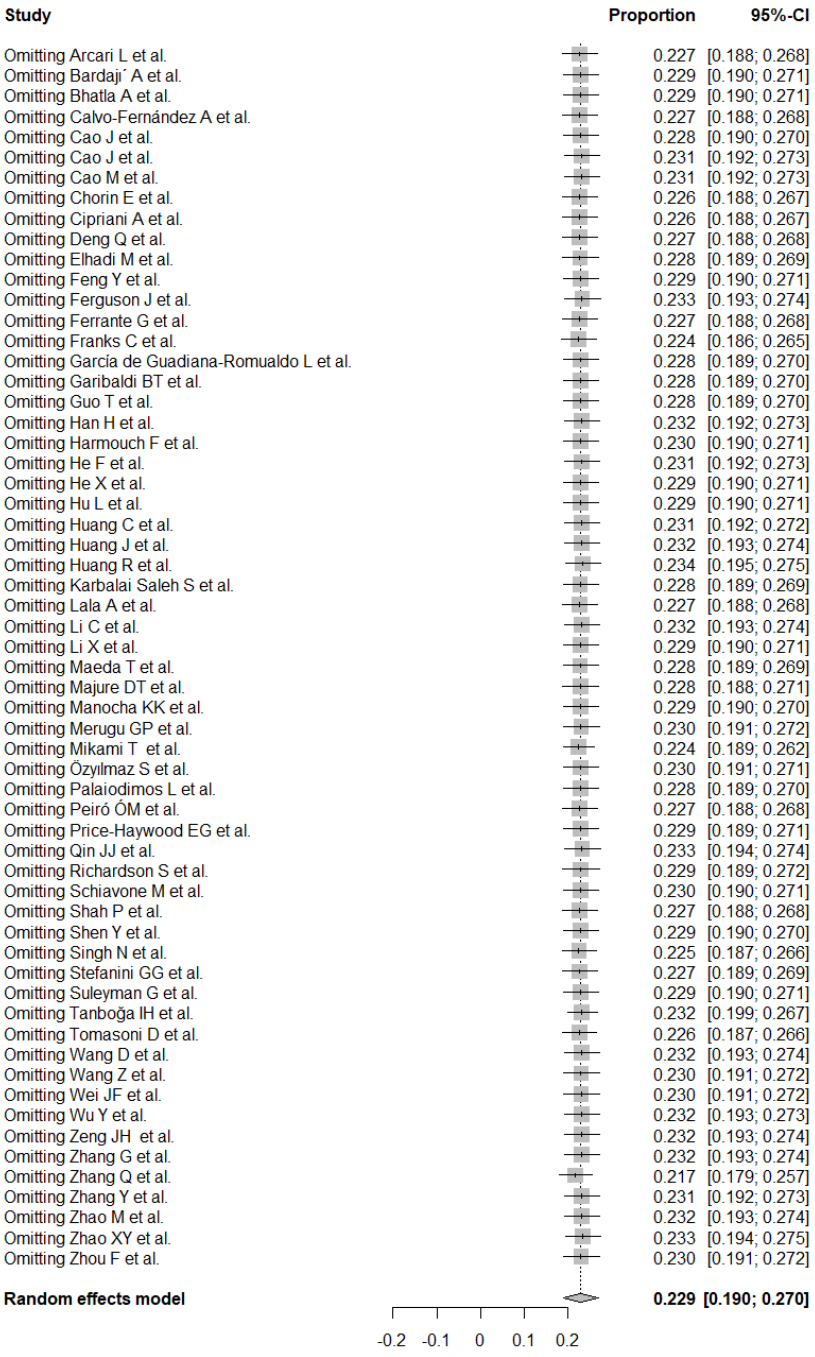

Study
Events
Total
Proportion
95%-CI
Weight (fixed)
Weight (random)
Bardaji´ A et al. 33 186 0.177 [0.125; 0.240] 0.6% 5.7%
Calvo-Fernández A et al. 58 872 0.067 [0.051; 0.085] 6.1% 6.1%
Cipriani A et al. 31 109 0.284 [0.202; 0.379] 0.2% 5.2%
García de Guadiana-Romualdo L et al. 249 1280 0.195 [0.173; 0.217] 3.6% 6.1%
Maeda T et al. 52 181 0.287 [0.223; 0.359] 0.4% 5.5%
Majure D et al. 1888 6247 0.302 [0.291; 0.314] 12.9% 6.1%
Manocha KK et al. 237 446 0.531 [0.484; 0.578] 0.8% 5.8%
Peiró ÓM et al. 34 196 0.173 [0.123; 0.234] 0.6% 5.7%
Saleh S et al. 76 386 0.197 [0.158; 0.240] 1.1% 5.9%
Schiavone M et al. 46 674 0.068 [0.050; 0.090] 4.6% 6.1%
Shah P et al. 110 309 0.356 [0.303; 0.412] 0.6% 5.7%
Singh N et al. 93 276 0.337 [0.281; 0.396] 0.5% 5.7%
Stefanini G et al. 65 397 0.164 [0.129; 0.204] 1.3% 6.0%
Tanboğa IH et al. 2020 14855 0.136 [0.131; 0.142] 55.2% 6.2%
Zeng J et al. 35 416 0.084 [0.059; 0.115] 2.4% 6.0%
Zhang Y et al. 7 166 0.042 [0.017; 0.085] 1.8% 6.0%
Zhao M et al. 63 1000 0.063 [0.049; 0.080] 7.4% 6.1%
Fixed effect model 27996 0.152 [0.148; 0.156] 100.0% --
Random effects model 0.201 [0.153; 0.249] -- 100.0%
Heterogeneity: I^2 = 99%, τ^2 = 0.0097, p < 0.01
0.1
0.2
0.3
0.4
0.5

Figure S2. Forest plot showing result of sensitivity analysis after excluding each study in turn

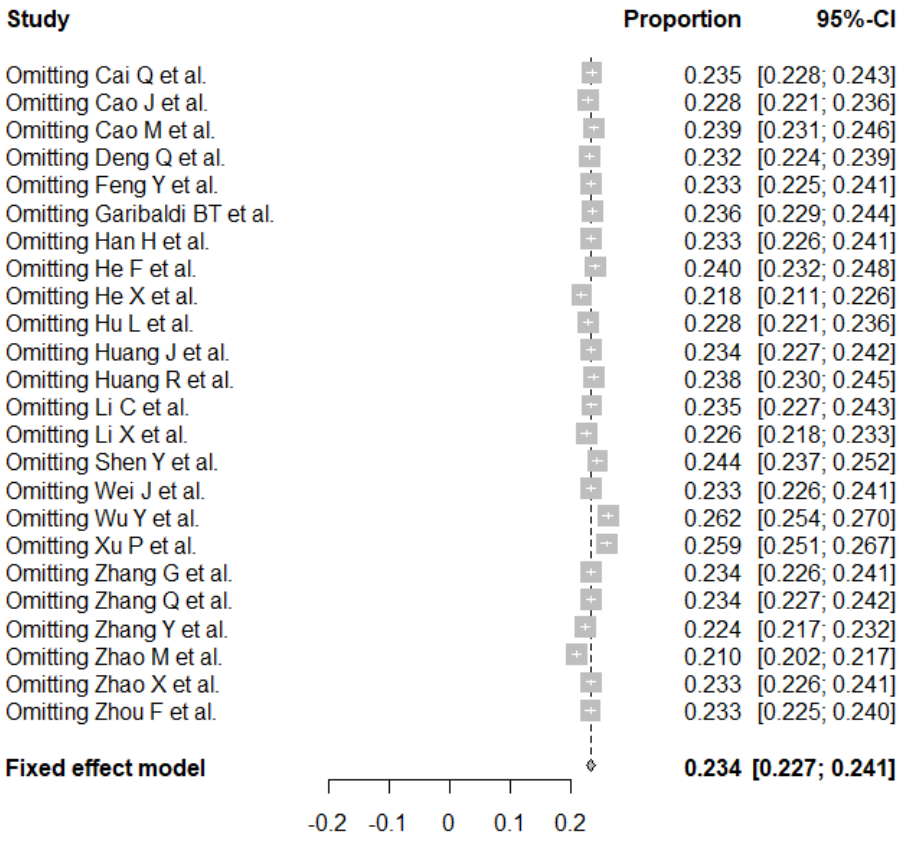
A. Sensitivity analysis of the prevalence of elevated Mb in the general population



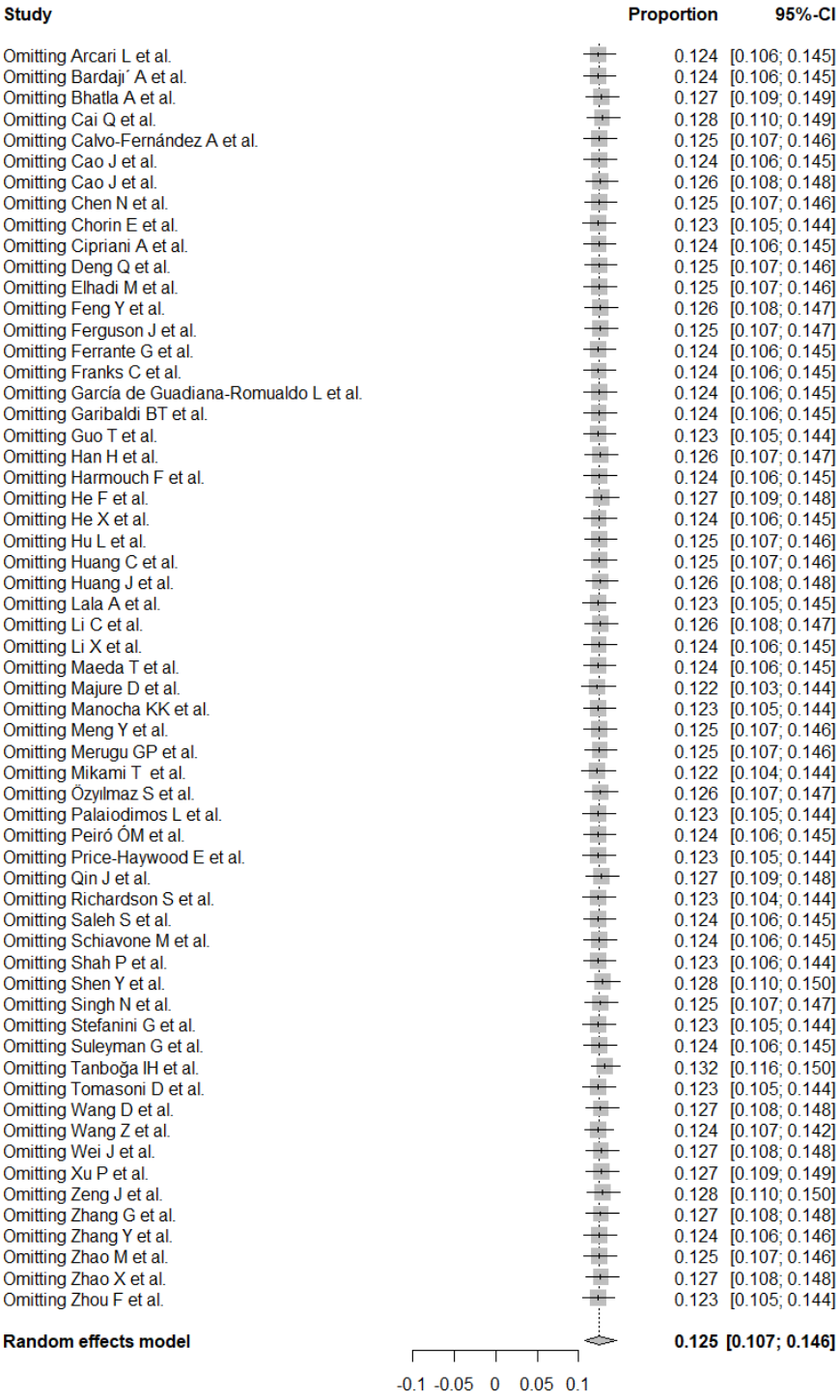
B. Sensitivity analysis of the prevalence of elevated cTnI in the general population



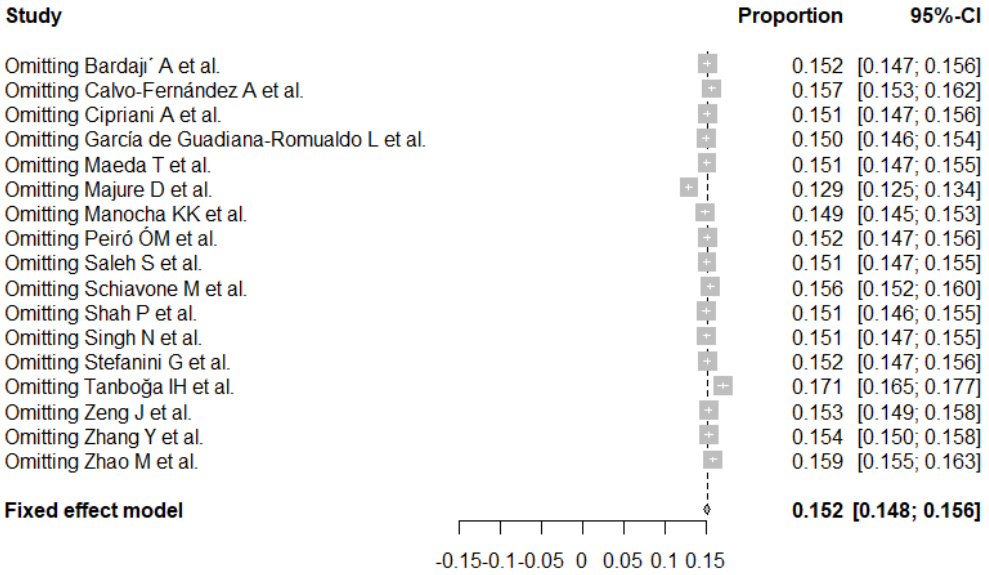
C. Sensitivity analysis of the incidence of severe illness



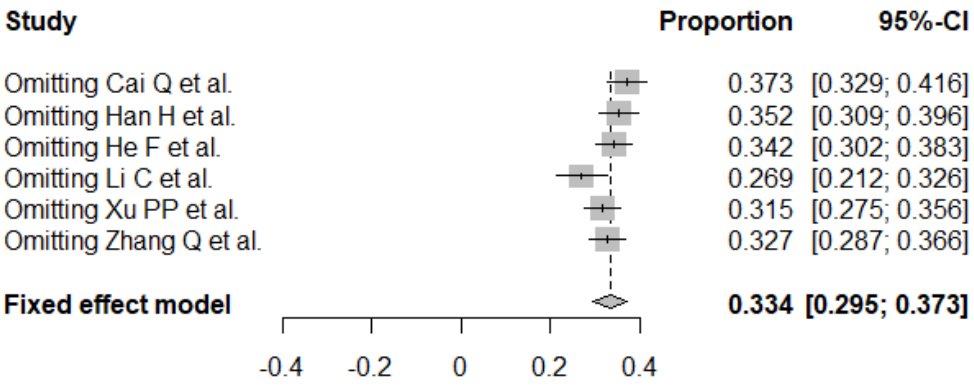
D. Sensitivity analysis of the incidence of in-hospital mortality



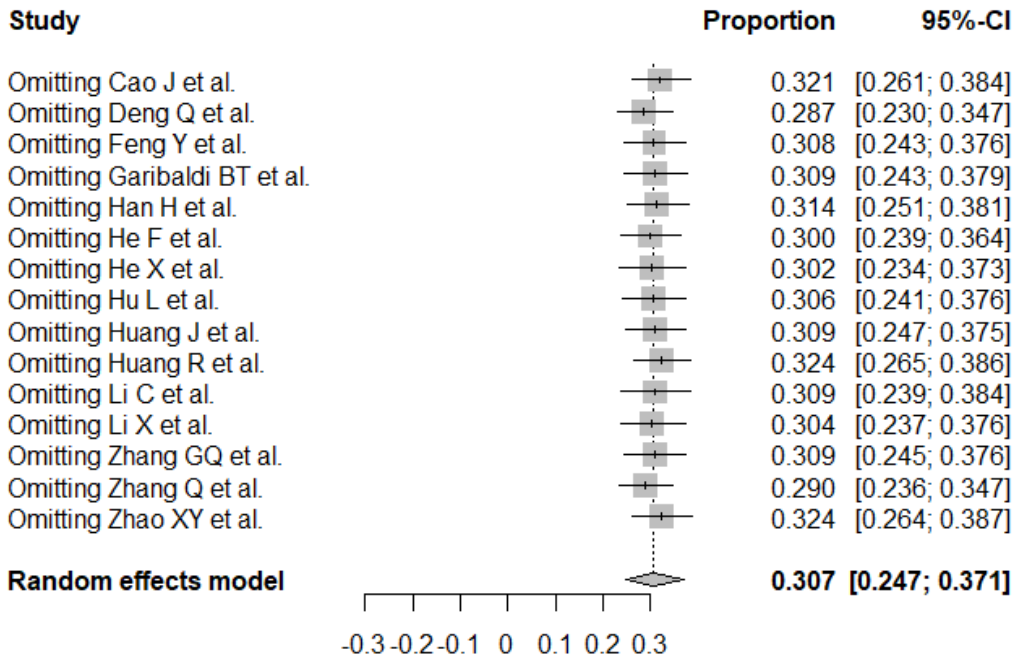
E. Sensitivity analysis of the incidence of ICU-admission



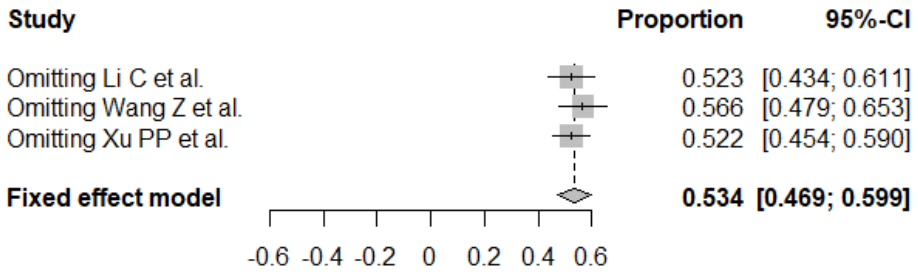
F. Sensitivity analysis of the prevalence of elevated Mb in severe disease group



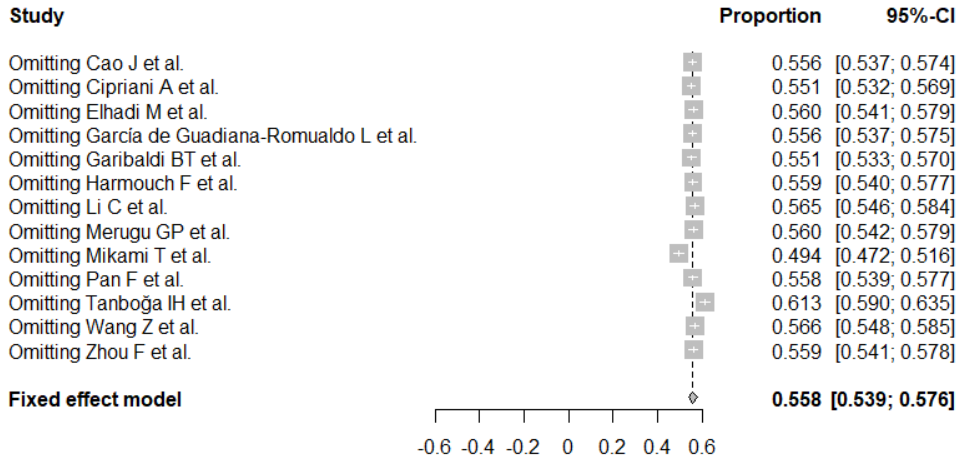
G. Sensitivity analysis of the prevalence of elevated cTnI in severe disease group



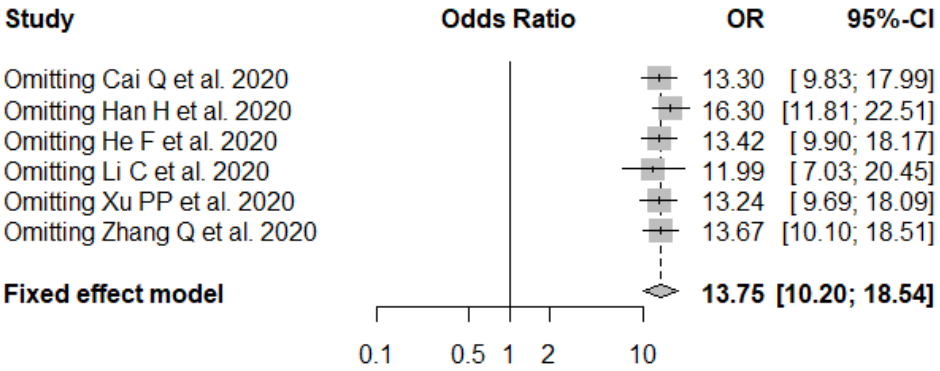
H. Sensitivity analysis of the prevalence of elevated Mb in non-survivor group



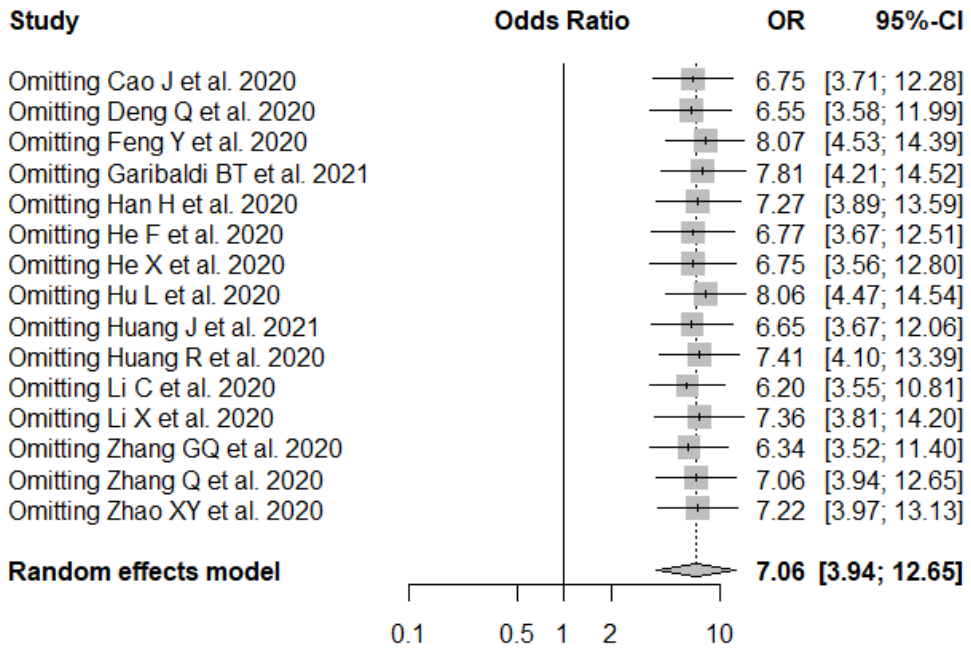
I. Sensitivity analysis of the prevalence of elevated cTnI in non-survivor group



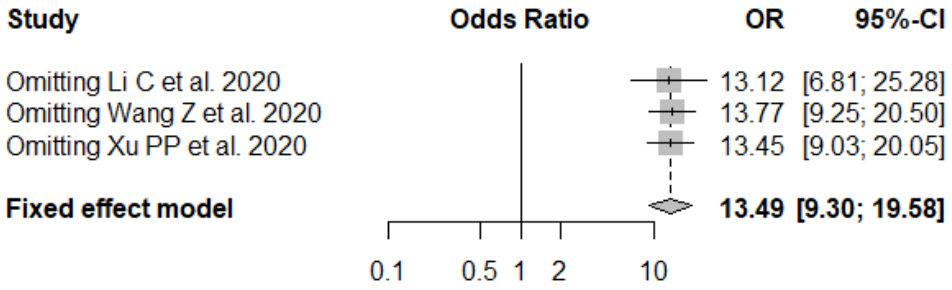
J. Sensitivity analysis of the risk of elevated Mb for severe illness



K. Sensitivity analysis of the risk of elevated cTnI for severe illness



L. Sensitivity analysis of the risk of elevated Mb for in-hospital mortality



M. Sensitivity analysis of the risk of elevated cTnI for in-hospital mortality

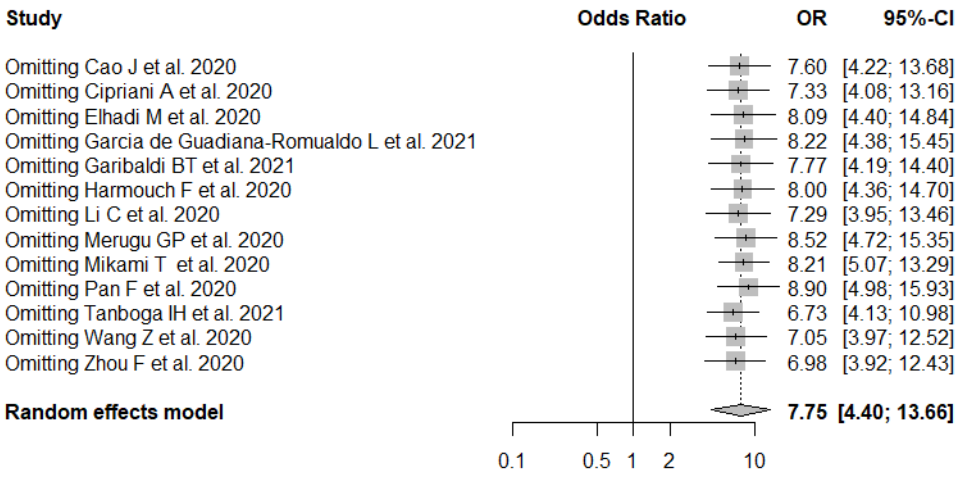
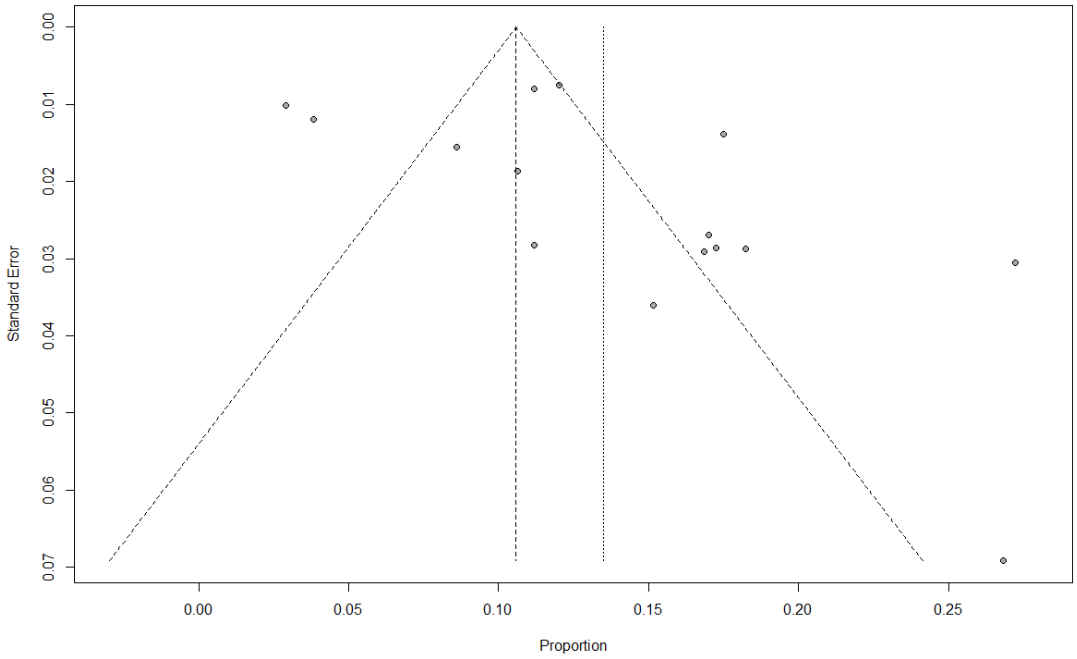


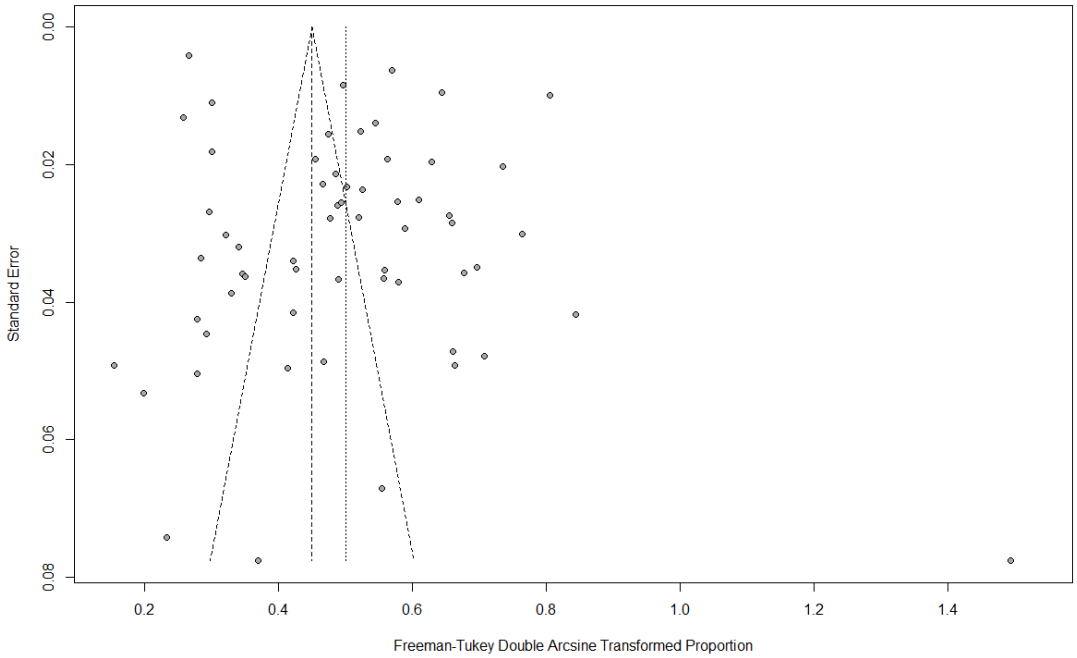
Figure S3. Funnel plot

A. Funnel plot of the prevalence of elevated Mb in the general population



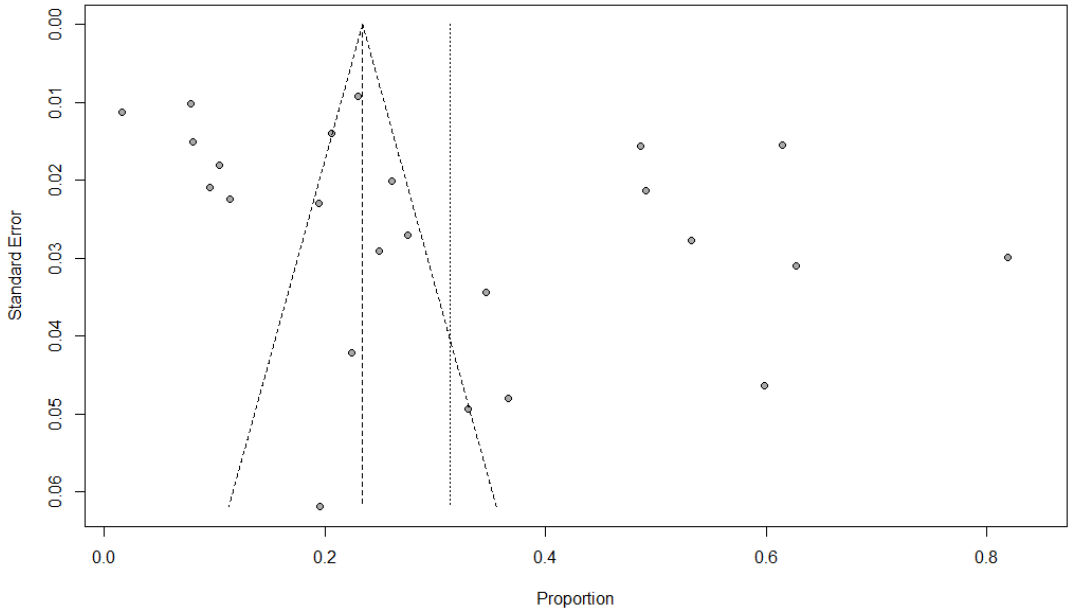
Egger p-value: 0.09453; Begger p-value: 0.08326; Peter p-value: 0.08192

B. Funnel plot of the prevalence of elevated cTnI in the general population



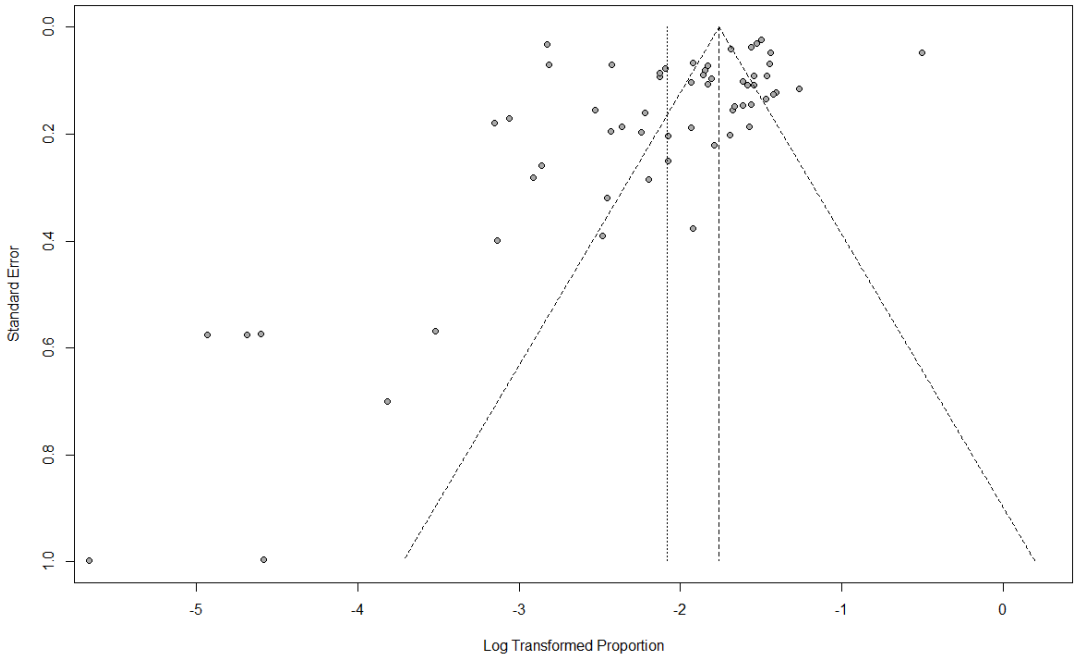
Egger p-value: 0.03789; Begger p-value: 0.07207; Peter p-value: 0.5637

C. Funnel plot of the incidence of severe illness



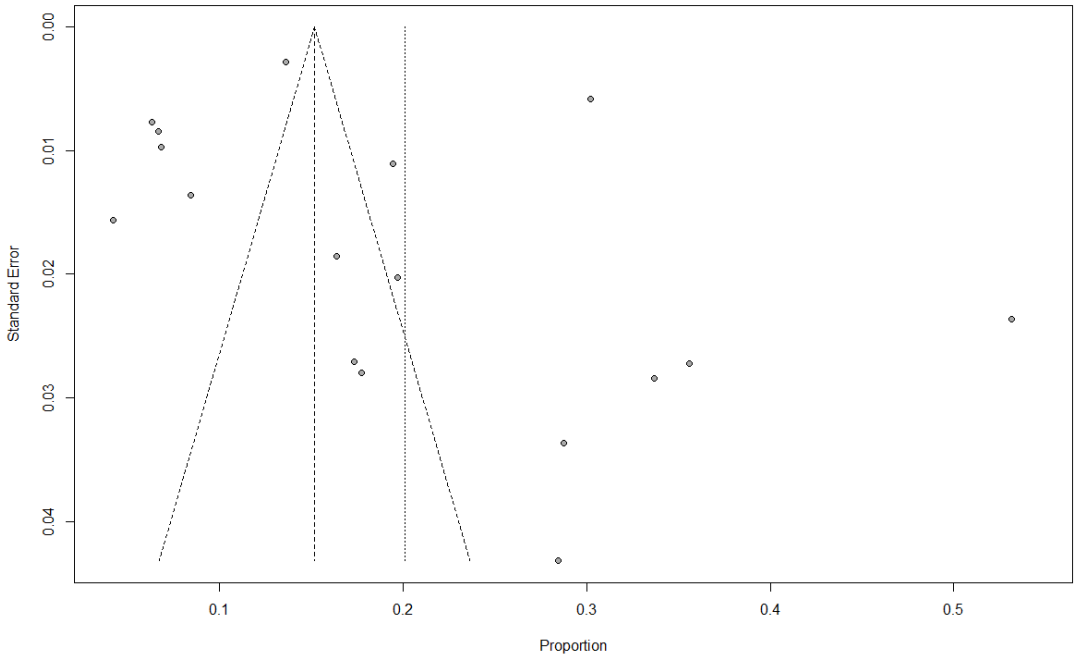
Egger p-value: 0.02466; Begger p-value: 0.01726; Peter p-value: 0.7713

D. Funnel plot of the incidence of in-hospital mortality



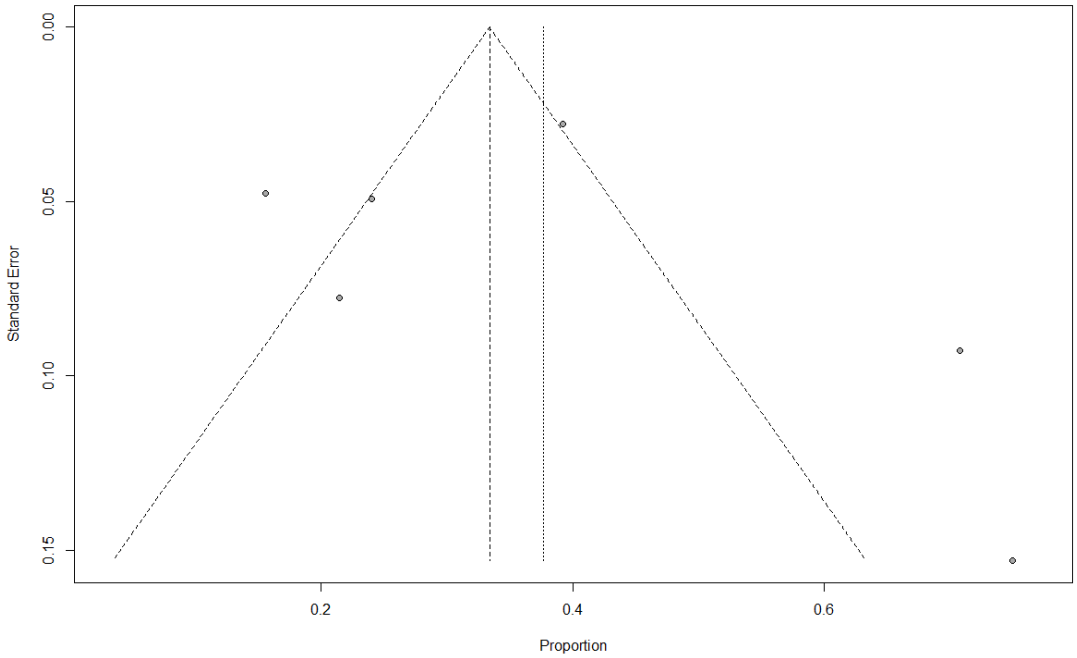
Egger p-value: 0.04792; Begger p-value: 0.0003063; Peter p-value: 0.4989

E. Funnel plot of the incidence of ICU-admission



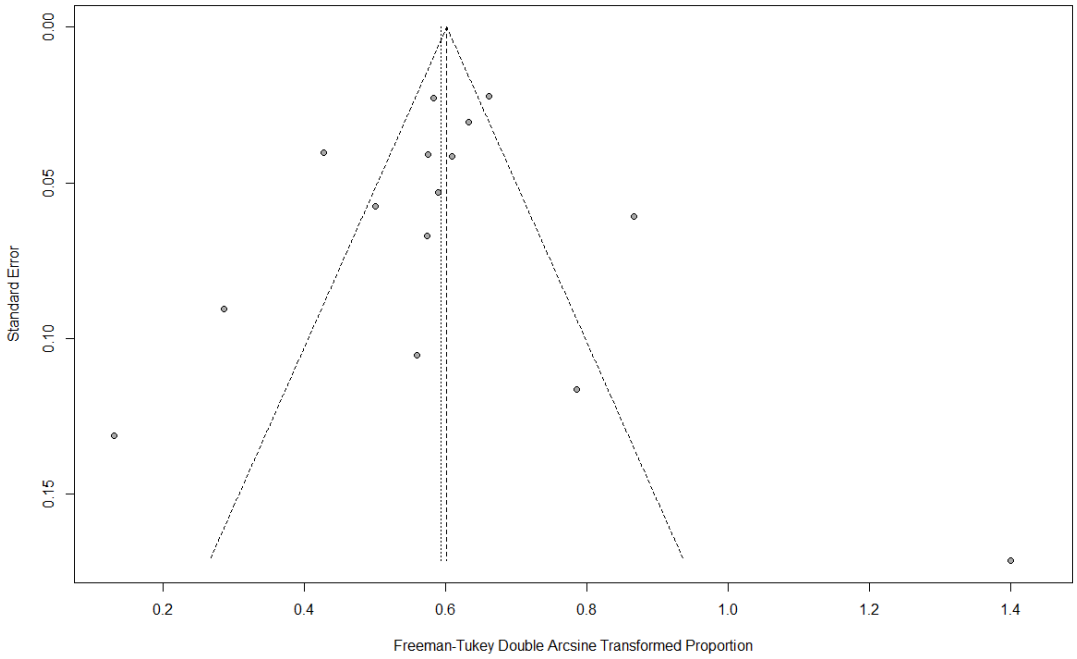
Egger p-value: 0.3967; Begger p-value: 0.02612; Peter p-value: 0.4039

F. Funnel plot of the prevalence of elevated Mb in severe disease group



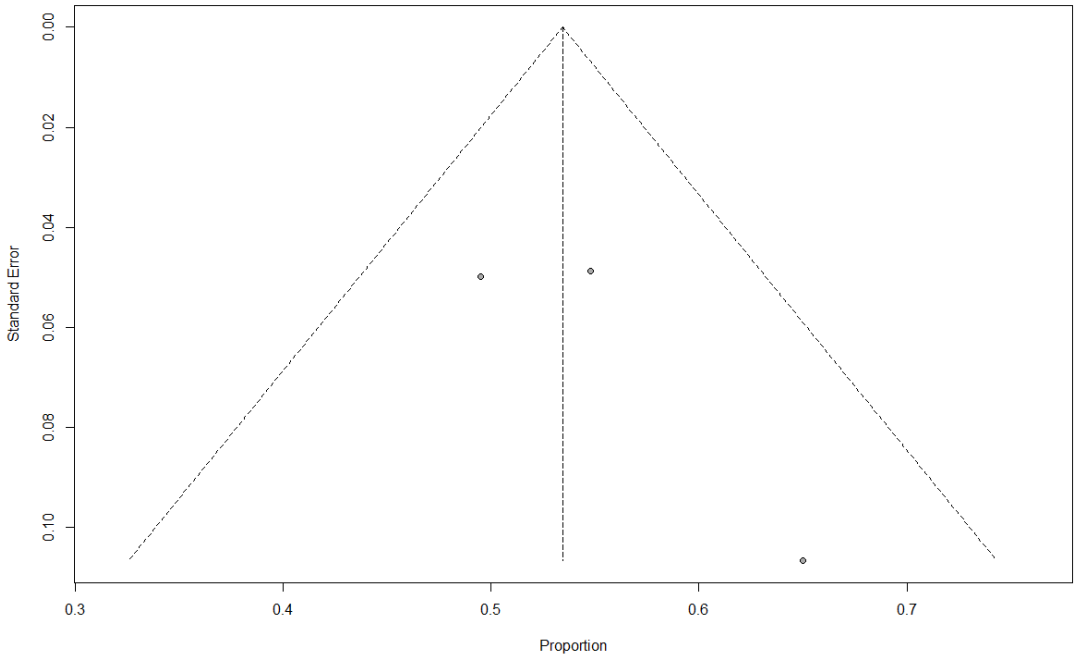
Egger p-value: 0.6935; Begger p-value: 0.3746; Peter p-value: 0.5328

G. Funnel plot of the prevalence of elevated cTnI in severe disease group



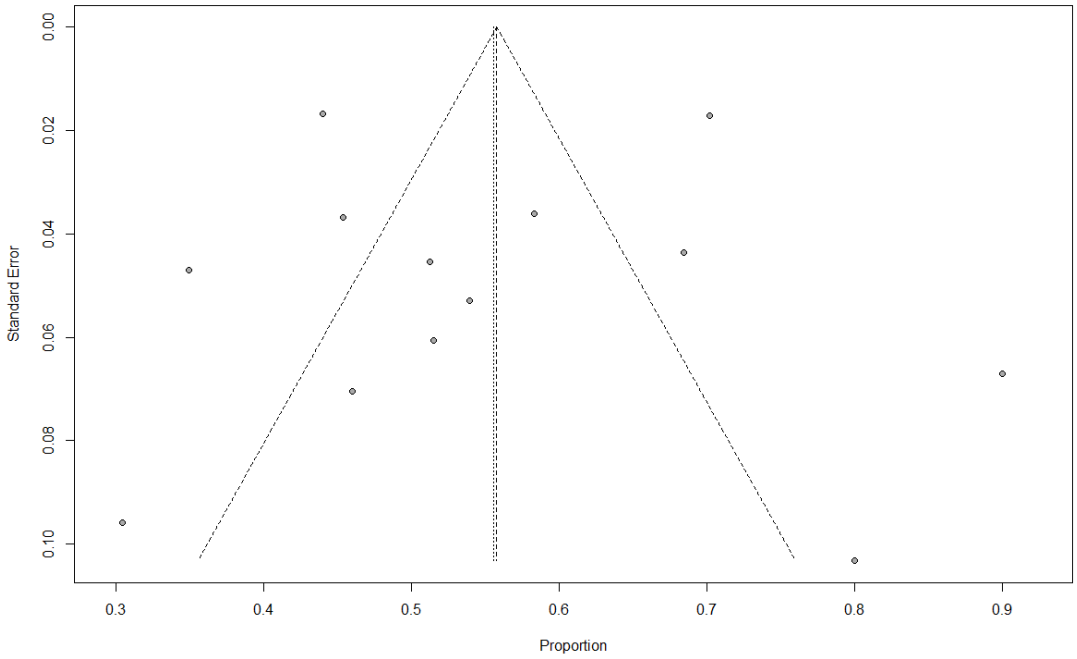
Egger p-value: 0.8258; Begger p-value: 0.882; Peter p-value: 0.8568

H. Funnel plot of the prevalence of elevated Mb in non-survivor group



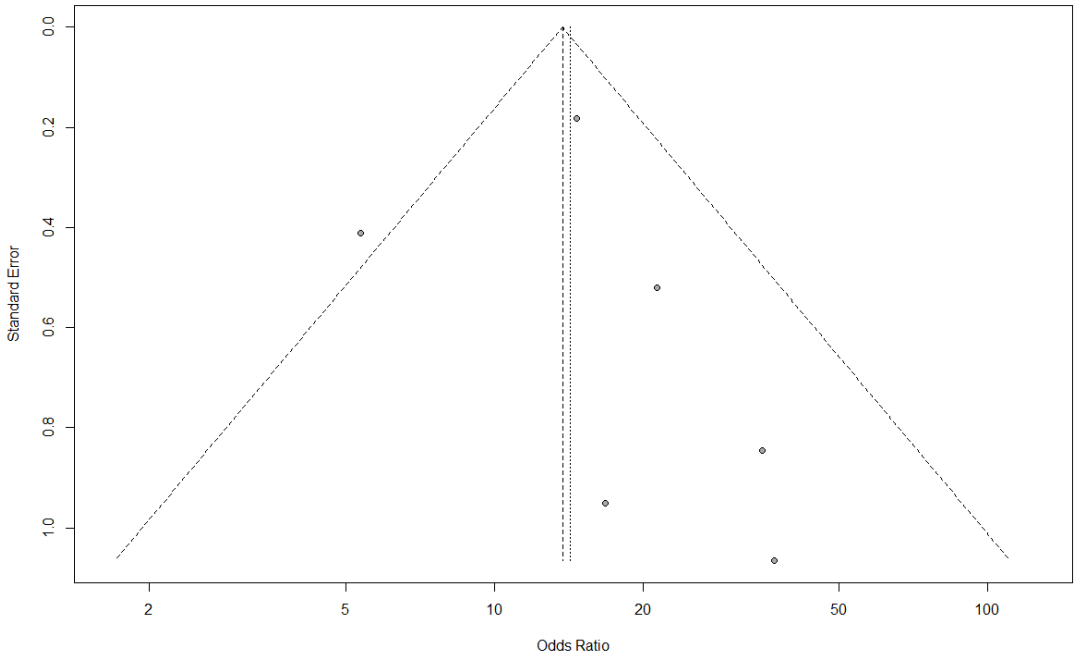
Egger p-value: 0.3916; Begger p-value: 0.6015; Peter p-value: 0.4063

I. Funnel plot of the prevalence of elevated cTnI in non-survivor group



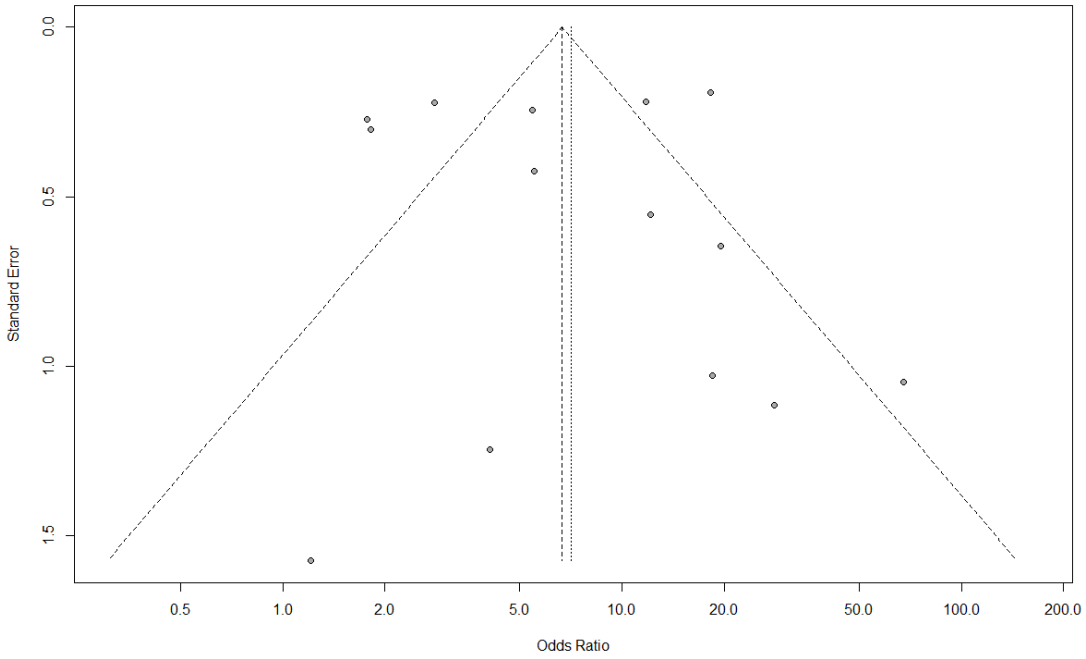
Egger p-value: 0.9162; Begger p-value: 0.9029; Peter p-value: 0.9438

J. Funnel plot of the risk of elevated Mb for severe illness



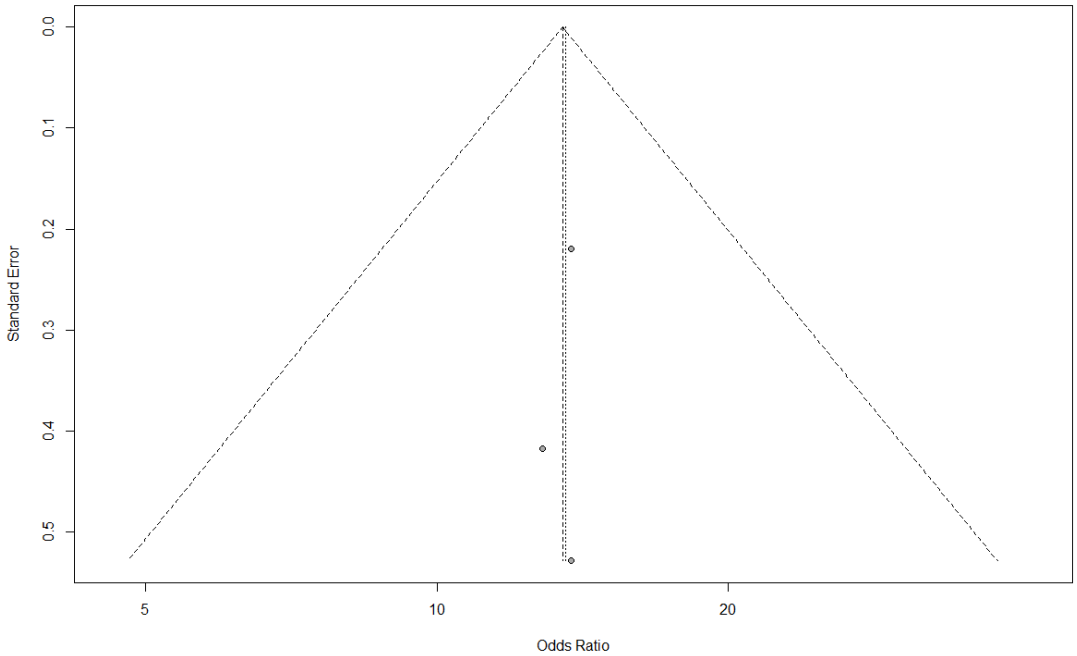
Egger p-value: 0.6346; Begger p-value: 0.3476; Peter p-value: 0.8624

K. Funnel plot of the risk of elevated cTnI for severe illness



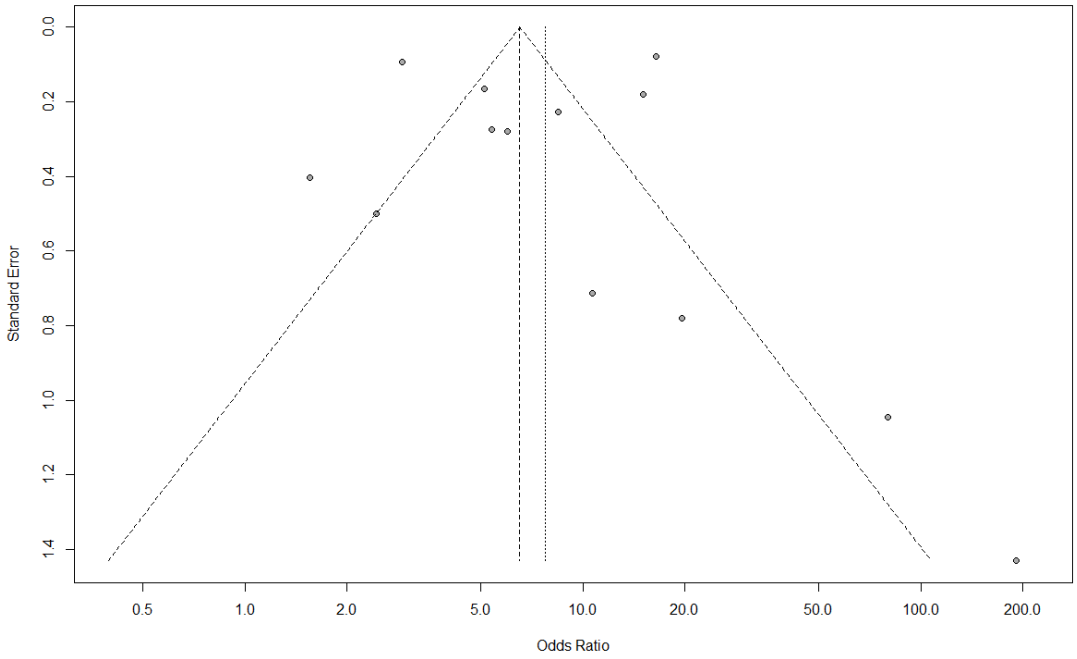
Egger p-value: 0.8127; Begger p-value: 0.8695; Peter p-value: 0.7997

L. Funnel plot of the risk of elevated Mb for in-hospital mortality



Egger p-value: 0.6612; Begger p-value: 0.6015; Peter p-value: 0.567

M. Funnel plot of the risk of elevated cTnI for in-hospital mortality



Egger p-value: 0.8723; Begger p-value: 0.2721; Peter p-value: 0.9568

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